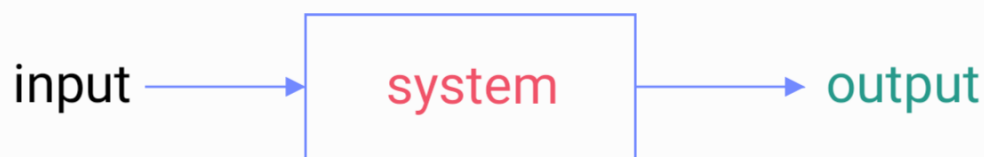


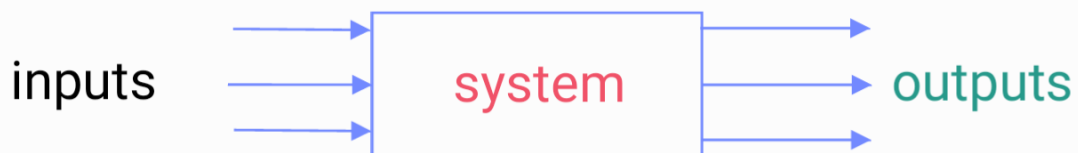
Control System

→ system: when a number of elements or components are connected in a sequence to perform a specific task. Then the arrangements thus formed is called as a system.

↳ ex: computer, human body, hospital, etc.



SISO: Single Input and Single Output system



MIMO: Multiple Input and Multiple Output system

→ control system: an arrangement of physical components connected in a proper manner to produce a desired output.

↳ the output quantity is controlled by varying the input quantity.

↳ ex: electric iron, air conditioner, etc.

→ Bio control system:

↳ the mammalian organism maintains stable, efficient and homeostasis in the face of external and internal perturbations via distinct biological systems ranging from the large-scale physiological (nervous, immune, endocrine, circulatory, respiratory, etc.), to the cellular (growth and proliferation regulation, DNA damage repair, etc.) and the subcellular (gene expression, protein synthesis, metabolite regulation, etc.).

↳ bio control systems, a sub-topic of control theory, arises from a control engineering perspective of the function, organization and combination of these multiscale biological systems and the control mechanisms that enable them to carryout their functions effectively.

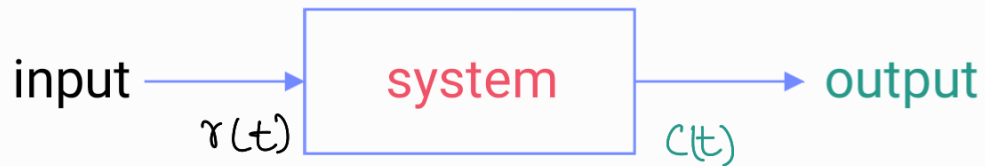
↳ ex: linear model of respiratory mechanics, linear model of lung mechanics, etc.

→ types of control system:

↳ open loop system

↳ closed loop system

→ open loop system:

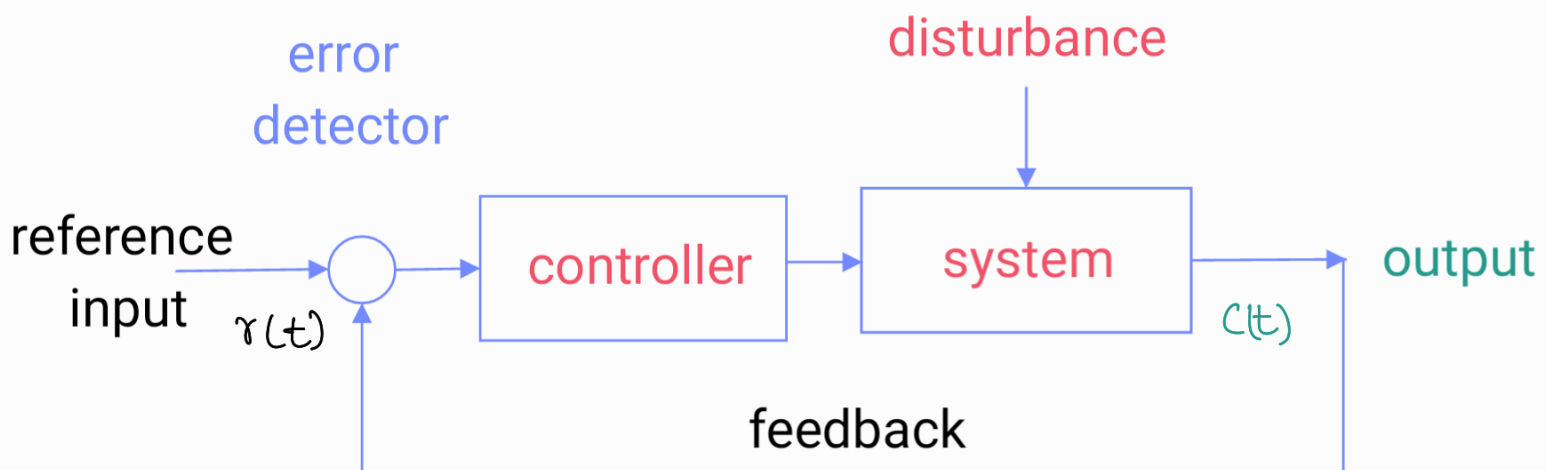


↳ output is directly controlled by changing the input manually.

↳ due to external disturbance $\Rightarrow$ system output may change.

↳ output is fed back to the input for correction.

→ closed loop system:



↳ output has an effect of upon the input in order to maintain the desired output.

↳ open loop system can be modified as closed loop system by providing feed back to the input.

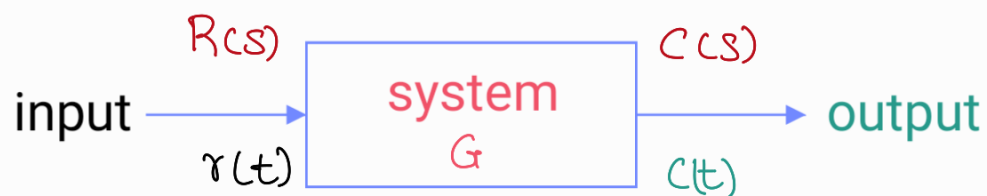
↳ provisions of feedback $\Rightarrow$ automatically corrects the system output changes due to disturbances.

- ↳ reference signal corresponds to desired output.
- ↳ feedback path elements samples the output.
- ↳ feedback signal is proportional to output signal
⇒ fed to the error detector.
- ↳ difference between reference signal and feedback signal ⇒ error signal.
- ↳ controller modifies and amplifies the error signal
⇒ better control action.
- ↳ modified error signal ⇒ fed to plant to correct output.

open loop system	closed loop system
→ simple and economical	→ complex and costly
→ easy to maintain	→ difficult to maintain
→ inaccurate	→ accurate
→ more sensitive	→ less sensitive
→ optimization is not possible	→ optimization is possible
→ generally stable	→ more care is needed to design a stable system
→ high gain	→ feedback reduces overall gain

→ transfer function:

- ↳ system can be represented by differential equations.
- ↳ transfer function can be obtained by taking Laplace transform of equations governing the system.
- ↳ ratio of the Laplace transform of the output to the Laplace transform of the input under the assumptions of all initial conditions as zero.
- ↳ characterize the input-output relationship of the system.



$$G(s) = \frac{C(s)}{R(s)}$$

$$T.F = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

→ feedback principles:

- ↳ positive feedback
- ↳ negative feedback

→ positive feedback:

- ↳ amplifies deviations from a desired state.
- ↳ used in systems where a rapid increase is desired such as oscillators.
- ↳ ex: microphone placed too close to a speaker ⇒ audio feedback loop ⇒ sound is amplified repeatedly.

→ negative feedback:

- ↳ reduces deviations from a desired state.
- ↳ used in most control systems and feedback amplifiers.
- ↳ ex: thermostat maintain the temperature at a set point by turning the heater on/off as needed.

→ characteristics of control system:

- ↳ stability: output remains bounded and does not oscillate or diverge uncontrollables in response to a disturbance or input.

ex: a stable cruise control system maintains a vehicle's speed without causing it to accelerate uncontrollably.

↳ accuracy: how closely the systems output matches the desired input or reference value.

ex: an automated factory robot positioning parts according to specifications to ensure quality.

↳ speed of response: how quickly a system respond to changes in input or disturbances.

ex: a drone adjusting its position when the pilot changes direction.

↳ robustness: systems ability to perform well despite uncertainties or variations in the enviroment.

ex: a self-driving car handling different road conditions and unexpected obstacles.

↳ sensitivity: how responsive a system is to changes in input or disturbances.

ex: small adjustments in frequency setting affects the sound output of an audio equalizer.

↳ transient response: how the system behaves during the transition from one state to another.

ex: voltage stabilizes after being switched on in an electronic circuit.

↳ steady -state error: difference between desired output and actual output once the system has settled and is operating in a steady state.

ex: temperature control system.

Mathematical Modeling of Physical Systems

→ control system $\Rightarrow$ a collection of physical components connected together to serve an objective.

→ input-output relations of components of a system are governed by differential equations.

→ mathematical model constitutes a set of differential equations.

→ response of the system can be studied by solving the differential equations for various input conditions.

→ linear time invariant $\Rightarrow$ coefficients of the differential equation are constants.

→ linear time variant $\Rightarrow$ coefficients of the differential equation are functions of time.

→ **mechanical systems:**

↳ translational mechanical systems: motion along a straight line.



↳ rotational mechanical systems: motion along its own axis.



Mechanical translational systems

- mathematical model can be obtained by using three basic elements mass, spring and damper.
- mass: weight of the mechanical system concentrated at centre of the body.
- spring: elastic deformation of the body.
- damper: friction of the body.
- when a force is applied to a translational system $\Rightarrow$ opposing force due to mass, friction and elasticity of the body $\Rightarrow$ governed by Newton's law of motion.
- net force acting on body is zero $\Rightarrow$ sum of applied forces is equal to sum of opposing forces on a body.
 - $\hookrightarrow x \rightarrow$ displacement (m)
 - $\hookrightarrow v = (dx/dt) \rightarrow$ velocity (m/s)
 - $\hookrightarrow a = (dv/dt) = (d^2x/dt^2) \rightarrow$ acceleration (m/s^2)
 - $\hookrightarrow f \rightarrow$ applied force (N)
 - $\hookrightarrow f_m \rightarrow$ opposing force offered by mass (N)
 - $\hookrightarrow f_k \rightarrow$ opposing force offered by elasticity (N)
 - $\hookrightarrow f_b \rightarrow$ opposing force offered by friction (N)
 - $\hookrightarrow M \rightarrow$ mass (kg)
 - $\hookrightarrow K \rightarrow$ stiffness of spring (N/m)
 - $\hookrightarrow B \rightarrow$ viscous friction coefficient (Ns/m)

→ ideal mass element:

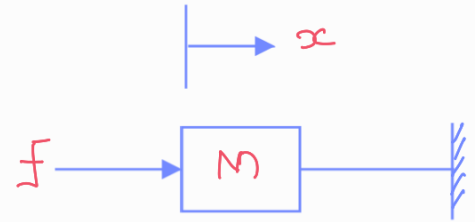
$$f \propto a$$

$$f = m a$$

$$= m \frac{dv}{dt}$$

$$f = m \frac{d^2x}{dt^2}$$

$$f = f_m = m \frac{d^2x}{dt^2}$$

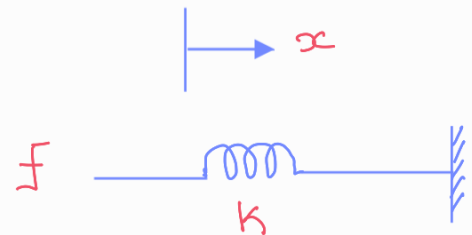


→ ideal spring element with one end fixed to reference:

$$f \propto x$$

$$f = kx$$

$$f = f_k = kx$$

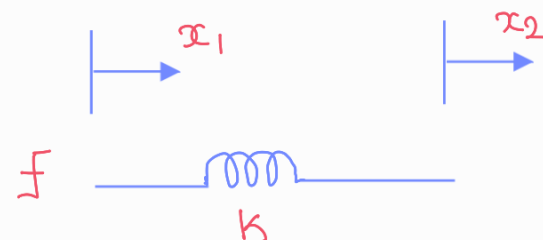


→ ideal spring element with displacement at both ends:

$$f \propto (x_1 - x_2)$$

$$f = k(x_1 - x_2)$$

$$f = f_k = k(x_1 - x_2)$$

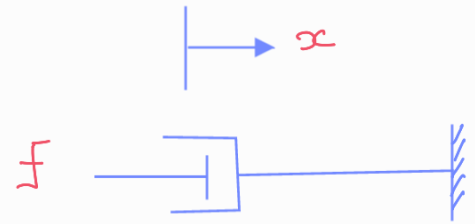


→ ideal damper element with one end fixed to reference:

$$f \propto \frac{dx}{dt}$$

$$f = B \frac{dx}{dt}$$

$$f = f_b = B \frac{dx}{dt}$$

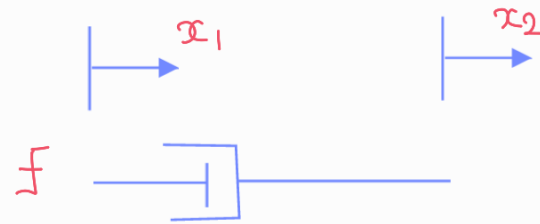


→ ideal damper element with displacement at both ends:

$$f \propto \frac{d}{dt}(x_1 - x_2)$$

$$f = B \frac{d}{dt}(x_1 - x_2)$$

$$f = f_b = B \frac{d}{dt}(x_1 - x_2)$$



→ transfer function of mechanical translational system:

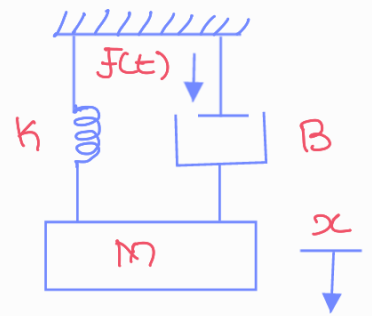
↳ obtain governing differential equations from force balance equations at nodes in the system.

↳ assume linear displacement of the nodes as $x_1, x_2, \dots$ and assign a displacement to each node (mass).

↳ obtain the free body diagram by drawing each node separately and then marking all the forces acting on that node.

↳ D'Alembert's principle : for any body, the algebraic sum of externally applied forces and forces resisting the motion in any given direction is zero.

↳ write the differential equation by equating the sum of applied forces to the sum of opposing forces.



↳ take Laplace transform of differential equations to convert them to algebraic equations.

$$F(t) = M \frac{d^2 x}{dt^2} + B \frac{dx}{dt} + kx$$

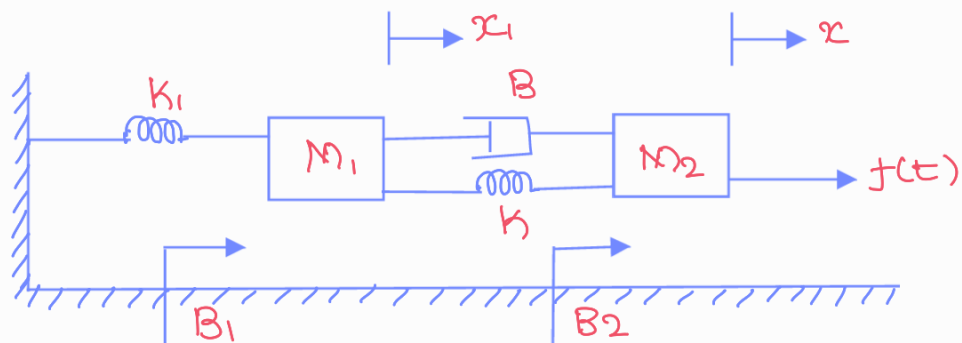
$$F(s) = Ms^2 X(s) + BsX(s) + kX(s)$$

↳ obtain the ratio between output variable and input variable.

$$F(s) = X(s) [Ms^2 + Bs + k]$$

$$\frac{X(s)}{F(s)} = \frac{1}{ms^2 + Bs + k}$$

1. write the differential equations governing the mechanical system and determine the transfer function.



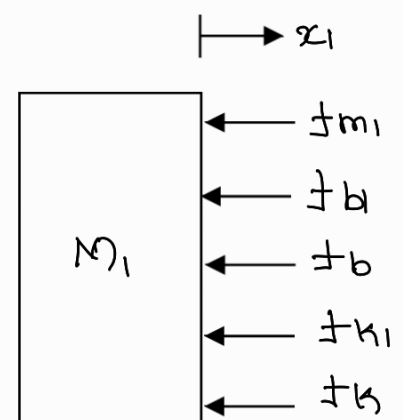
node 1: $f_{m1} + f_{b1} + f_b + f_{k1} + f_k = 0$

differential equation:

$$M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B \frac{d(x_1 - x_2)}{dt} + k_1 x_1 + k(x_1 - x_2) = 0$$

Laplace transform:

$$M_1 s^2 X_1(s) + B_1 s X_1(s) + Bs [X_1(s) - X_2(s)] + k_1 X_1(s) + k [X_1(s) - X_2(s)] = 0$$



$$M_1 s^2 X_1(s) + B_1 s X_1(s) + B s X_1(s) - B s X(s) + k_1 X_1(s) + k X_1(s) - k X(s) = 0$$

$$X_1(s) [M_1 s^2 + B_1 s + B s + k_1 + k] - X(s) [B s + k] = 0$$

$$X_1(s) [M_1 s^2 + (B_1 + B)s + (k_1 + k)] - X(s) [B s + k] = 0$$

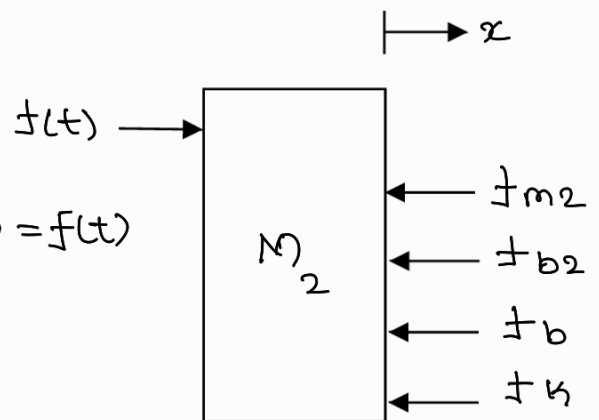
$$X_1(s) [M_1 s^2 + (B_1 + B)s + (k_1 + k)] = X(s) [B s + k]$$

$$X_1(s) = \frac{X(s) [B s + k]}{M_1 s^2 + (B_1 + B)s + (k_1 + k)}$$

node 2: $f_{m2} + f_{b2} + f_b + f_k = f(t)$

differential equation:

$$M_2 \frac{d^2 x}{dt^2} + B_2 \frac{dx}{dt} + B \frac{d(x - x_1)}{dt} + k(x - x_1) = f(t)$$



Laplace transform:

$$M_2 s^2 X(s) + B_2 s X(s) + B s [X(s) - X_1(s)] + k [X(s) - X_1(s)] = F(s)$$

$$M_2 s^2 X(s) + B_2 s X(s) + B s X(s) - B s X_1(s) + k X(s) - k X_1(s) = F(s)$$

$$X(s) [M_2 s^2 + B_2 s + B s + k] - X_1(s) [B s + k] = F(s)$$

$$X(s) [M_2 s^2 + (B_2 + B)s + k] - X_1(s) [B s + k] = F(s)$$

$$X(s) [M_2 s^2 + (B_2 + B)s + k] - \frac{X(s) [B s + k] [B s + k]}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} = F(s)$$

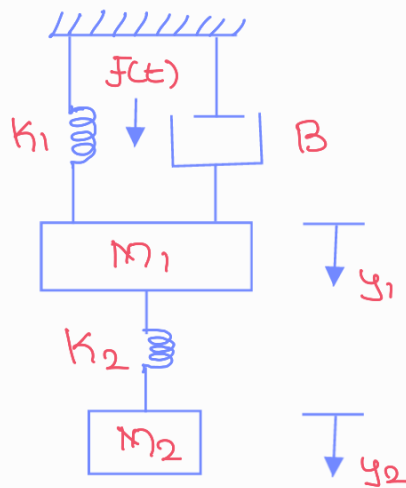
$$X(s) \left[\frac{M_2 s^2 + (B_2 + B)s + k - [B s + k]^2}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} \right] = F(s)$$

$$X(s) \left\{ \frac{[M_2 s^2 + (B_2 + B)s + k] [M_1 s^2 + (B_1 + B)s + (k_1 + k)]}{M_1 s^2 + (B_1 + B)s + (k_1 + k)} \right\} - (Bs + k)^2 = F(s)$$

transfer function:

$$\frac{X(s)}{F(s)} = \frac{M_1 s^2 + (B_1 + B)s + (k_1 + k)}{\{ [M_2 s^2 + (B_2 + B)s + k] [M_1 s^2 + (B_1 + B)s + (k_1 + k)] \} - (Bs + k)^2}$$

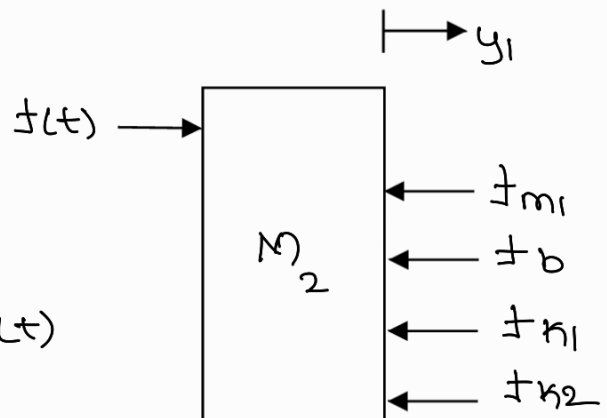
2. Determine the transfer function of the system.



node 1: $f_{m_1} + f_b + f_{k_1} + f_{k_2} = f(t)$

differential equation:

$$m_1 \frac{d^2 y_1}{dt^2} + B \frac{dy_1}{dt} + k_1 y_1 + k_2 (y_1 - y_2) = f(t)$$



Laplace transform:

$$m_1 s^2 Y_1(s) + B s Y_1(s) + k_2 [Y_1(s) - Y_2(s)] = F(s)$$

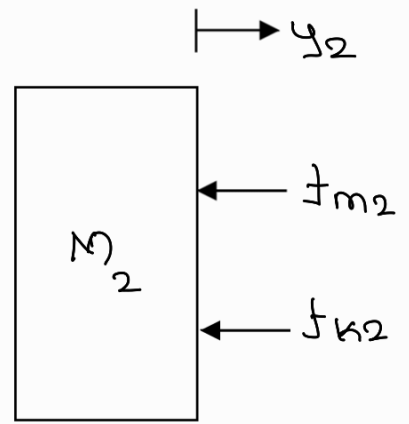
$$Y_1(s) [M_1 s^2 + B s + k_2] - Y_2(s) k_2 = F(s)$$

node 2:

$$f_{m_2} + f_{k_2} = 0$$

differential equation:

$$m_2 \frac{d^2 y_2}{dt^2} + k_2 (y_2 - y_1) = 0$$



Laplace transform:

$$m_2 s^2 Y_2(s) + k_2 [Y_2(s) - Y_1(s)] = 0$$

$$Y_2(s) [m_2 s^2 + k_2] - Y_1(s) k_2 = 0$$

$$Y_1(s) k_2 = Y_2(s) [m_2 s^2 + k_2]$$

$$Y_1(s) = Y_2(s) \frac{[m_2 s^2 + k_2]}{k_2}$$

$$Y_1(s) [m_1 s^2 + B s + k_2] - Y_2(s) k_2 = F(s)$$

$$Y_2(s) \left[\frac{[m_2 s^2 + k_2]}{k_2} [m_1 s^2 + B s + k_2] - Y_2(s) k_2 \right] = F(s)$$

$$Y_2(s) \frac{\{[m_2 s^2 + k_2] [m_1 s^2 + B s + k_2] - k_2^2\}}{k_2} = F(s)$$

transfer function:

$$\frac{Y_2(s)}{F(s)} = \frac{k_2}{\{[m_2 s^2 + k_2] [m_1 s^2 + B s + k_2] - k_2^2\}}$$

Mechanical rotational systems

→ mathematical model can be obtained by using three elements moment of inertia of mass, torsional spring with stiffness and damper with rotational frictional coefficient.

→ moment of inertia of mass: weight of the rotational mechanical system concentrated at the centre of gravity of the body.

→ torsional spring with stiffness: elastic deformation of the body.

→ damper with rotational frictional coefficient: friction existing in rotational mechanical system.

→ when a torque is applied to a rotational system $\Rightarrow$ opposing force due to moment of inertia, friction and elasticity of the system $\Rightarrow$ governed by Newton's law.

→ net torque acting on body is zero $\Rightarrow$ sum of applied torques is equal to sum of opposing torques on a body.

↳ $\theta \rightarrow$ angular displacement (rad)

↳ $(d\theta/dt) \rightarrow$ angular velocity (rad/s)

↳ $(d^2\theta/dt^2) \rightarrow$ angular acceleration (rad/s²)

↳ $T \rightarrow$ applied torque (N-m)

↳ $J \rightarrow$ moment of inertia (kg-m²/rad)

↳ $B \rightarrow$ rotational frictional coefficient (N-m-s/rad)

↳ $K \rightarrow$ stiffness of the torsional spring (N-m/rad)

→ ideal rotational mass element:

opposing torque due to moment of inertia

$$T_j \propto \frac{d^2\theta}{dt^2}$$

$$T_j = J \frac{d^2\theta}{dt^2}$$

$$T = T_j = J \frac{d^2\theta}{dt^2}$$



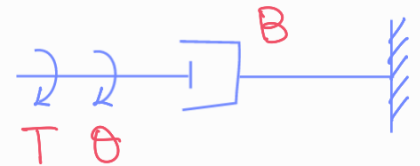
→ ideal rotational damper with one end fixed to reference:

opposing torque due to friction

$$T_b \propto \frac{d\theta}{dt}$$

$$T_b = B \frac{d\theta}{dt}$$

$$T = T_b = B \frac{d\theta}{dt}$$



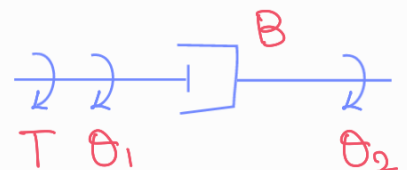
→ ideal rotational damper with angular displacement at both ends:

opposing torque due to friction

$$T_b \propto \frac{d(\theta_1 - \theta_2)}{dt}$$

$$T_b = B \frac{d(\theta_1 - \theta_2)}{dt}$$

$$T = T_b = B \frac{d(\theta_1 - \theta_2)}{dt}$$



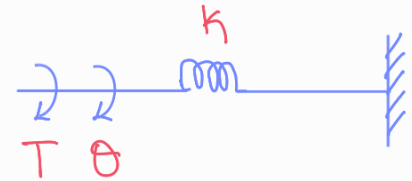
→ ideal rotational torsional spring with one end fixed to reference:

opposing torque due to elasticity

$$T_k \propto \theta$$

$$T_k = k \theta$$

$$T = T_k = k \theta$$



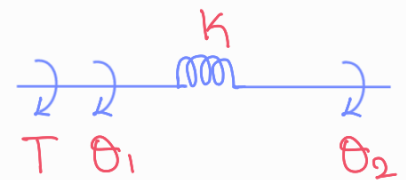
→ ideal rotational torsional spring with angular displacement at both ends:

opposing torque due to elasticity

$$T_k \propto (\theta_1 - \theta_2)$$

$$T_k = k(\theta_1 - \theta_2)$$

$$T = T_k = k(\theta_1 - \theta_2)$$



→ transfer function of mechanical translational system:

↳ obtain governing differential equations from torque balance equations at nodes in the system.

↳ assume angular displacement of the nodes as $\theta_1, \theta_2 \dots$ and assign a displacement to each node (mass).

↳ obtain the free body diagram by drawing each node separately and then marking all the torques acting on that node.

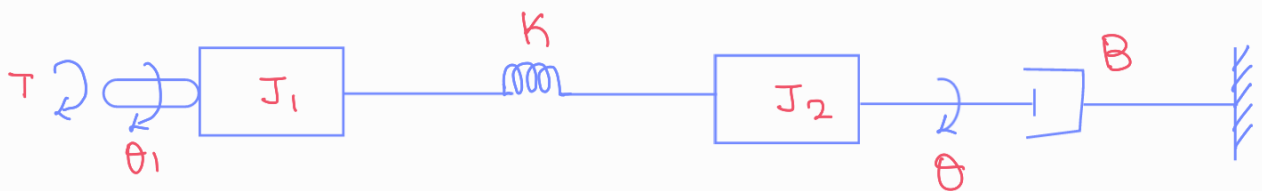
- ↳ write the differential equation by equating the sum of applied torques to the sum of opposing torques.
- ↳ take Laplace transform of differential equations to convert them to algebraic equations.
- ↳ obtain the ratio between output variable and input variable.

$$\mathcal{L} \{ \theta \} = \theta(s)$$

$$\mathcal{L} \left\{ \frac{d\theta}{dt} \right\} = s \theta(s)$$

$$\mathcal{L} \left\{ \frac{d^2\theta}{dt^2} \right\} = s^2 \theta(s)$$

3. write the differential equations governing the mechanical rotational system and obtain the transfer function of the system.

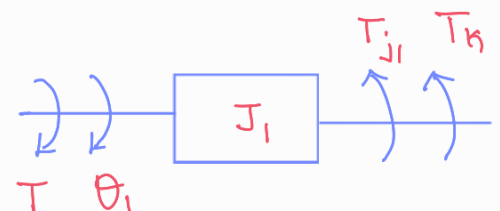


node 1:

$$T_{J1} + T_k = T$$

differential equations:

$$J_1 \frac{d^2\theta_1}{dt^2} + k(\theta_1 - \theta) = T$$



Laplace transform:

$$J_1 s^2 \theta_1(s) + k [\theta_1(s) - \theta(s)] = T(s)$$

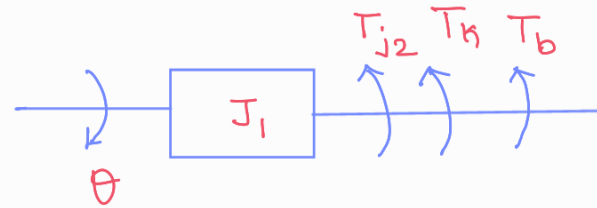
$$J_1 s^2 \theta_1(s) + k \theta_1(s) - k \theta(s) = T(s)$$

$$\theta_1(s) [J_1 s^2 + k] - k \theta(s) = T(s)$$

node 2:

$$T_{j_2} + T_b + T_k = 0$$

differential equations:



$$J_2 \frac{d^2 \theta}{dt^2} + B \frac{d\theta}{dt} + k(\theta - \theta_1) = 0$$

Laplace transform:

$$J_2 s^2 \theta(s) + B s \theta(s) + k [\theta(s) - \theta_1(s)] = 0$$

$$J_2 s^2 \theta(s) + B s \theta(s) + k \theta(s) - k \theta_1(s) = 0$$

$$\theta(s) [J_2 s^2 + B s + k] - k \theta_1(s) = 0$$

$$\theta(s) [J_2 s^2 + B s + k] = k \theta_1(s)$$

$$\theta_1(s) = \frac{\theta(s) [J_2 s^2 + B s + k]}{k}$$

$$\theta_1(s) [J_1 s^2 + k] - k \theta(s) = T(s)$$

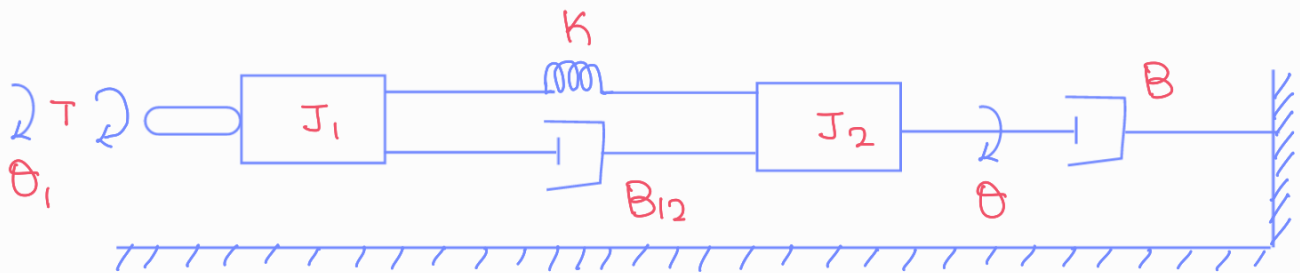
$$\frac{\theta(s) [J_2 s^2 + B s + k] [J_1 s^2 + k] - k \theta(s)}{k} = T(s)$$

$$\frac{\{ [J_1 s^2 + k] [J_2 s^2 + B s + k] - k^2 \} \theta(s)}{k} = T(s)$$

transfer function:

$$\frac{\theta(s)}{T(s)} = \frac{k}{[J_1 s^2 + k] [J_2 s^2 + B s + k] - k^2}$$

4. write the differential equations governing the mechanical rotational system and determine the transfer function.

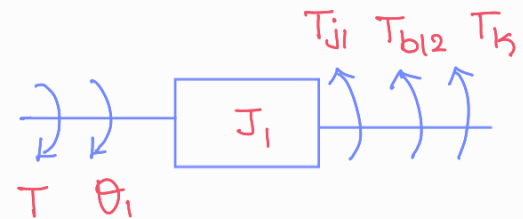


node 1:

$$T_{J1} + T_{B12} + T_k = T$$

differential equations:

$$J_1 \frac{d^2 \theta_1}{dt^2} + B_{12} \frac{d(\theta_1 - \theta)}{dt} + k(\theta_1 - \theta) = T$$



Laplace transform:

$$J_1 s^2 \theta_1(s) + B_{12} s [\theta_1(s) - \theta(s)] + k [\theta_1(s) - \theta(s)] = T(s)$$

$$J_1 s^2 \theta_1(s) + B_{12} s \theta_1(s) - B_{12} s \theta(s) + k \theta_1(s) - k \theta(s) = T(s)$$

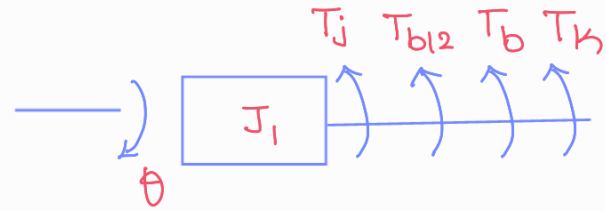
$$\theta_1(s) [J_1 s^2 + B_{12} s + k] - \theta(s) [B_{12} s + k] = T(s)$$

node 2:

$$T_j + T_{b12} + T_b + T_k = 0$$

differential equations:

$$J_2 \frac{d^2 \theta}{dt^2} + B_{12} \frac{d(\theta - \theta_1)}{dt} + B \frac{d\theta}{dt} + k(\theta - \theta_1) = 0$$



Laplace transform:

$$J_2 s^2 \theta(s) + B_{12} s [\theta(s) - \theta_1(s)] + B s \theta(s) + k [\theta(s) - \theta_1(s)] = 0$$

$$J_2 s^2 \theta(s) + B_{12} s \theta(s) - B_{12} s \theta_1(s) + B s \theta(s) + k \theta(s) - k \theta_1(s) = 0$$

$$\theta(s) [J_2 s^2 + B_{12} s + B s + k] - \theta_1(s) [B_{12} s + k] = 0$$

$$\theta(s) [J_2 s^2 + (B_{12} + B) s + k] - \theta_1(s) [B_{12} s + k] = 0$$

$$\theta(s) [J_2 s^2 + (B_{12} + B) s + k] = \theta_1(s) [B_{12} s + k]$$

$$\theta_1(s) = \frac{[J_2 s^2 + (B_{12} + B) s + k] \theta(s)}{[B_{12} s + k]}$$

$$\theta_1(s) [J_1 s^2 + B_{12} s + k] - \theta(s) [B_{12} s + k] = T(s)$$

$$\frac{[J_2 s^2 + (B_{12} + B) s + k] [J_1 s^2 + B_{12} s + k] - \theta(s) [B_{12} s + k]}{[B_{12} s + k]} = T(s)$$

$$\frac{[J_2 s^2 + (B_{12} + B) s + k] [J_1 s^2 + B_{12} s + k] - [B_{12} s + k]^2}{[B_{12} s + k]} \theta(s) = T(s)$$

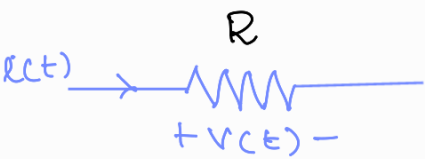
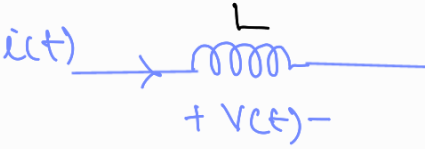
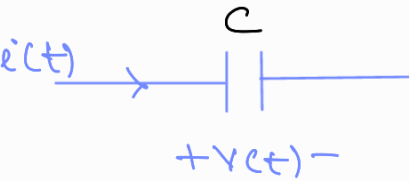
transfer function:

$$\frac{\theta(s)}{T(s)} = \frac{[B_{12} s + k]}{[J_2 s^2 + (B_{12} + B) s + k] [J_1 s^2 + B_{12} s + k] - [B_{12} s + k]^2}$$

Electrical systems

→ mathematical model can be obtained by using three elements resistor, capacitor and inductor.

→ current-voltage relation of R, L and C:

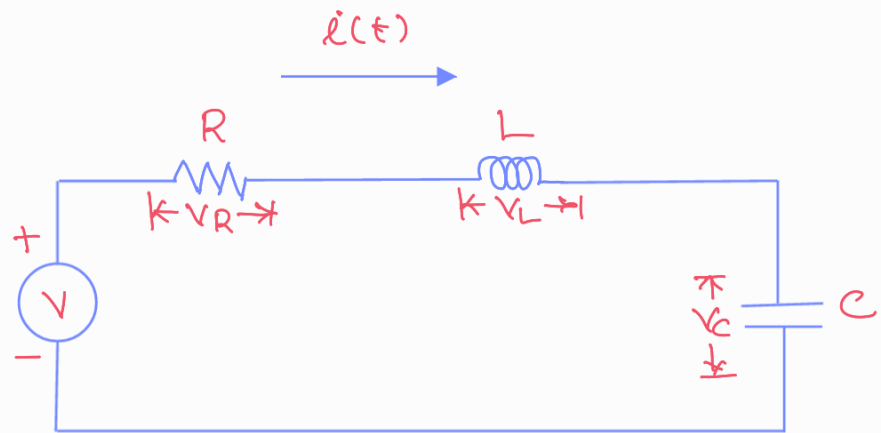
element	voltage across the element	current through the element
	$v(t) = R i(t)$	$i(t) = \frac{v(t)}{R}$
	$v(t) = L \frac{d i(t)}{dt}$	$i(t) = \frac{1}{L} \int v(t) dt$
	$v(t) = \frac{1}{C} \int i(t) dt$	$i(t) = C \frac{d v(t)}{dt}$

→ differential equations governing the electrical system can be formed by writing kirchoff's Current Law (KCL) equations by choosing various nodes in the network or kirchoff's Voltage Law (KVL) equations by choosing various closed paths in the network.

→ transfer function can be obtained by taking Laplace transform of differential equations and rearranging them as a ratio of output to input.

$$q = C v(t) ; \quad i(t) = \frac{dq}{dt} ; \quad v(t) = \frac{d\phi}{dt}$$

series combination of RLC:



→ applying KVL:

$$V = V_R + V_L + V_C$$

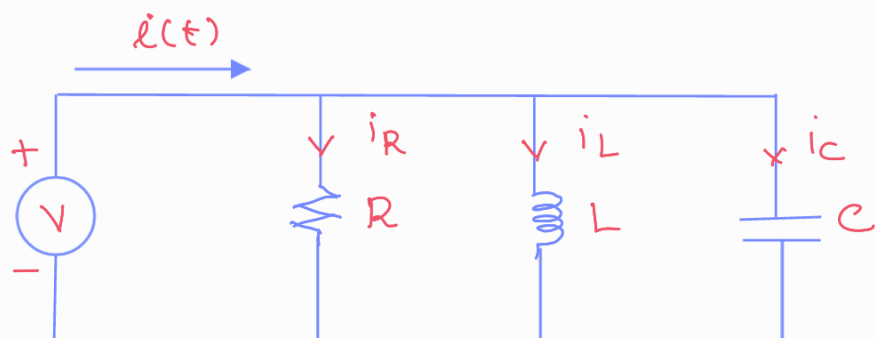
$$V = R i(t) + L \frac{d i(t)}{dt} + \frac{1}{C} \int i(t) dt$$

$$v = R \frac{dq}{dt} + L \frac{d^2 q}{dt^2} + \frac{1}{C} \int \frac{dq}{dt} dt$$

$$v = R \frac{dq}{dt} + L \frac{d^2 q}{dt^2} + \frac{1}{C} q$$

$$v = L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q$$

series combination of RLC:



→ applying KCL:

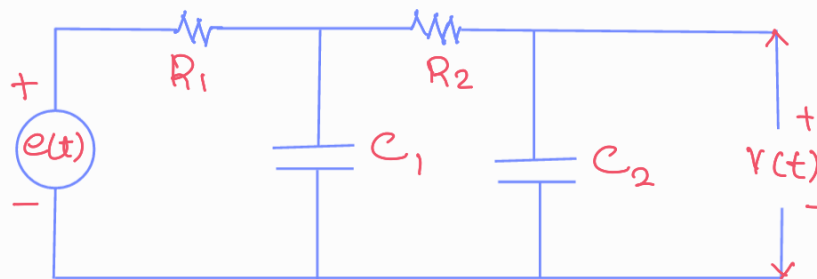
$$i = i_R + i_L + i_C$$

$$= \frac{v}{R} + \frac{1}{L} \int v dt + C \frac{dv}{dt}$$

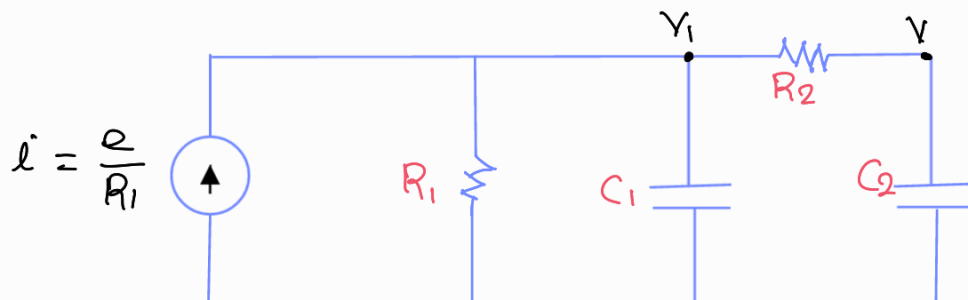
$$= \frac{1}{R} \frac{d\phi}{dt} + \frac{1}{L} \int \frac{d\phi}{dt} dt + C \frac{d^2\phi}{dt^2}$$

$$i = C \frac{d^2\phi}{dt^2} + \frac{1}{R} \frac{d\phi}{dt} + \frac{1}{L} \phi$$

5. Obtain the transfer function of the given electrical network.



→ voltage source with series resistor $\Rightarrow$ current source with parallel resistor



→ at node 1, applying KCL: $i = i_{R_1} + i_{C_1} + i_{R_2}$

$$\frac{e}{R_1} = \frac{v_1}{R_1} + C_1 \frac{dv_1}{dt} + \frac{v_1 - v}{R_2}$$

→ Laplace transform:
$$\frac{E(s)}{R_1} = \frac{V_1(s)}{R_1} + C_1 s V_1(s) + \frac{V_1(s) - V(s)}{R_2}$$

$$\frac{E(s)}{R_1} = V_1(s) \left[\frac{1}{R_1} + \frac{1}{R_2} + C_1 s \right] - \frac{V(s)}{R_2}$$

→ at node 2, applying KCL:

$$i_{R_2} = i_{C_2}$$

$$\frac{V_1 - V}{R_2} = C_2 \frac{dV}{dt}$$

→ Laplace transform:

$$\frac{V_1(s) - V(s)}{R_2} = C_2 s V(s)$$

$$V_1(s) - V(s) = C_2 s V(s) R_2$$

$$V_1(s) = C_2 s V(s) R_2 + V(s)$$

$$V_1(s) = [1 + R_2 C_2 s] V(s)$$

$$\frac{E(s)}{R_1} = V_1(s) \left[\frac{1}{R_1} + \frac{1}{R_2} + C_1 s \right] - \frac{V(s)}{R_2}$$

$$\frac{E(s)}{R_1} = [1 + R_2 C_2 s] V(s) \left[\frac{1}{R_1} + \frac{1}{R_2} + C_1 s \right] - \frac{V(s)}{R_2}$$

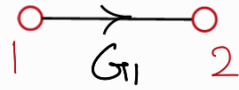
$$= [1 + s R_2 C_2] \left[\frac{R_2 + R_1 + R_1 R_2 C_1 s}{R_1 R_2} \right] V(s) - \frac{V(s)}{R_2}$$

$$\frac{E(s)}{R_1} = \frac{[[1 + s R_2 C_2] [R_2 + R_1 + R_1 R_2 C_1 s] - R_1] V(s)}{R_1 R_2}$$

Transfer function:
$$\frac{V(s)}{E(s)} = \frac{R_2}{[(1 + s R_2 C_2) (R_1 + R_2 + s R_1 R_2 C_1) - R_1]}$$

Signal Flow Graph

- Represent the control system graphically.
- **Node**: a point representing a variable/signal.



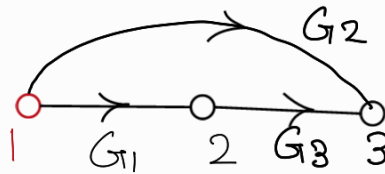
- **Branch**: directed line segment joining two nodes.



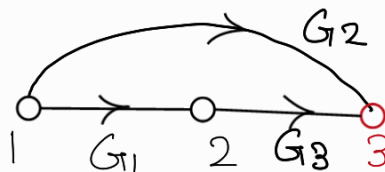
- **Transmittance**: gain of the branch.



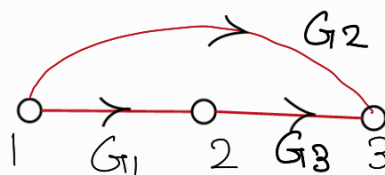
- **Input node**: node that has only outgoing branches.



- **Output node**: node that has only incoming branches.



- **Path**: traversal of connected branches in the direction of the branch arrows.



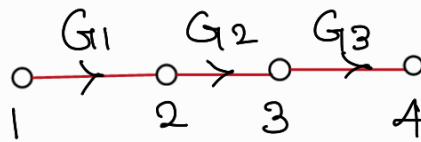
- **Open path**: Starts at a node and ends at another node.



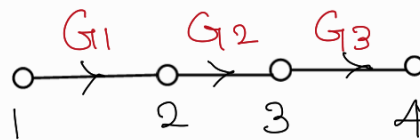
→ **Closed path:** Starts and ends at same node.



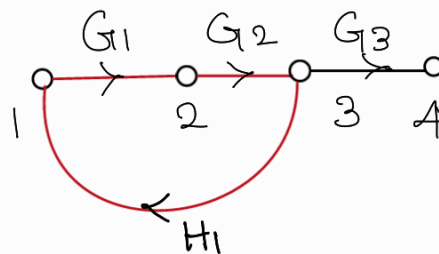
→ **Forward path:** a path from an input node to an output node that does not cross any node more than once.



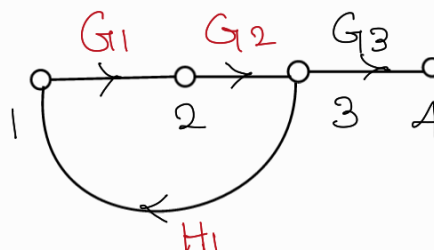
→ **Forward path gain:** the product of branch transmittances (gains) of a forward path.



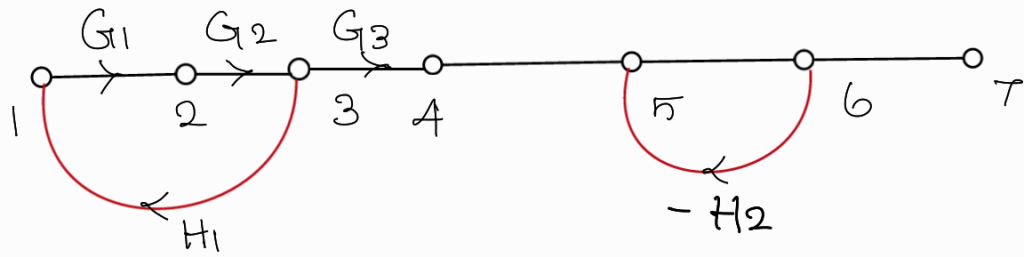
→ **Individual loop:** a closed loop path starting from a node and after passing through a Certain part of a graph arrives at same node without crossing any node more than once.



→ **Loop gain:** product of The branch transmittances (gains) of a loop.

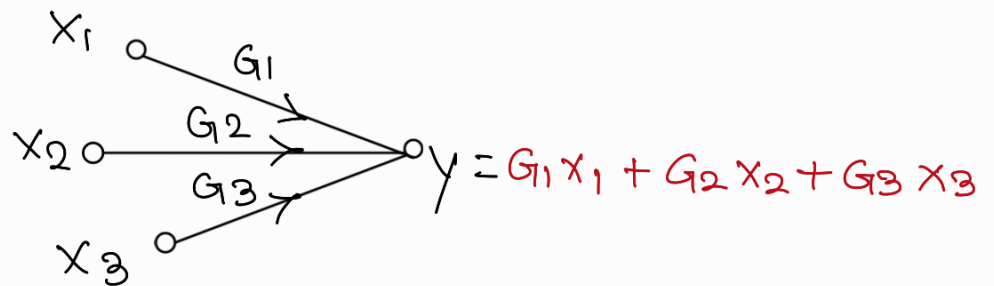


→ **Non-touching Loops:** If the loop does not have a common mode.

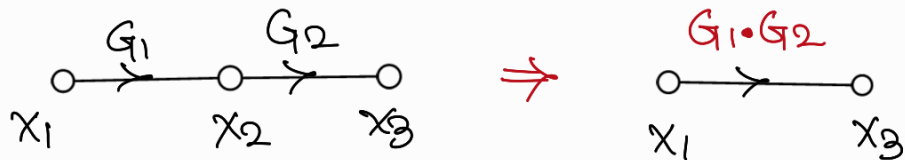


→ **Algebra:**

↳ Incoming signals:



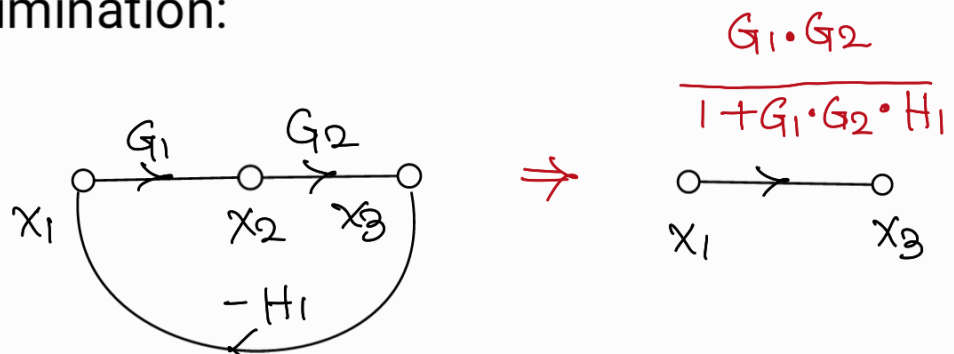
↳ Series connection:



↳ Parallel connection:



↳ Loop elimination:



→ **Signal flow graph Reduction:** either by using the rules of a signal Flow graph algebra or by using Mason's gain Formula.

→ Mason's Gain Formula:

$$T = \frac{\sum_{k=1}^n P_k \Delta_k}{\Delta}$$

T → transfer function of the system

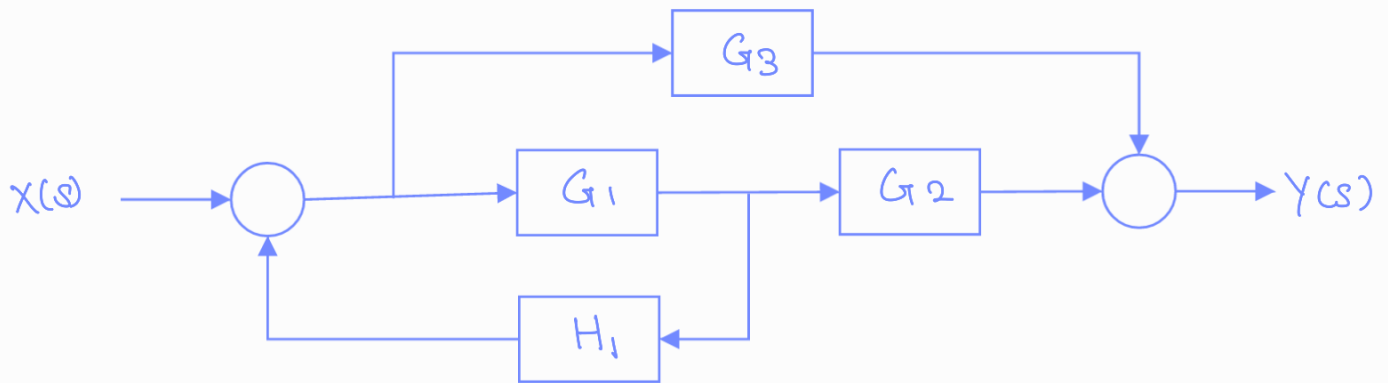
P → forward path gain

k → number of forward paths

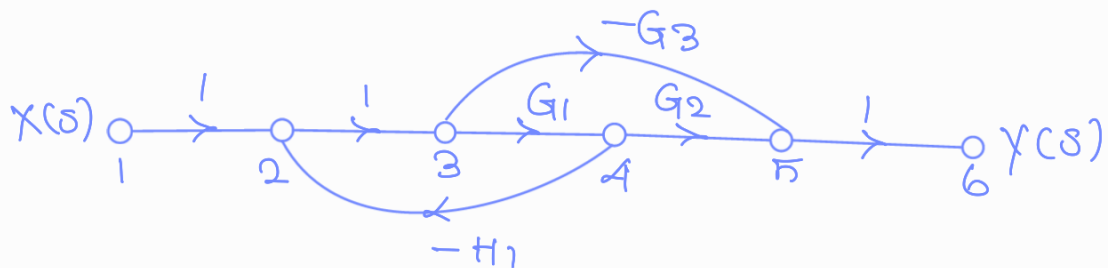
Δ → $1 - (\text{sum of individual loop gains}) +$
(sum of gain products of all possible combinations of
two non-touching loops) -
(sum of gain products of all possible combinations of
three non-touching loops) + ...

Δ_k → Δ for that part of the graph which is not touching
 k_{th} Forward path

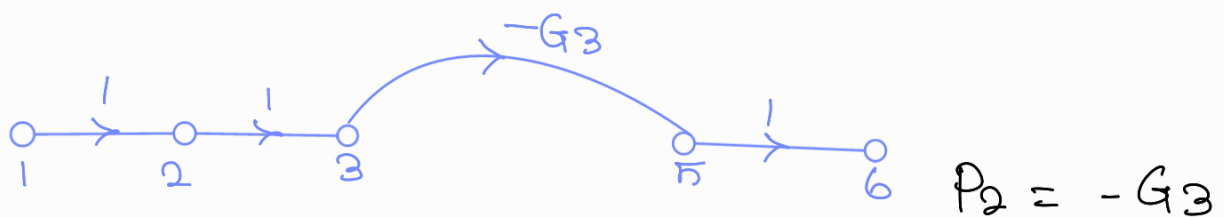
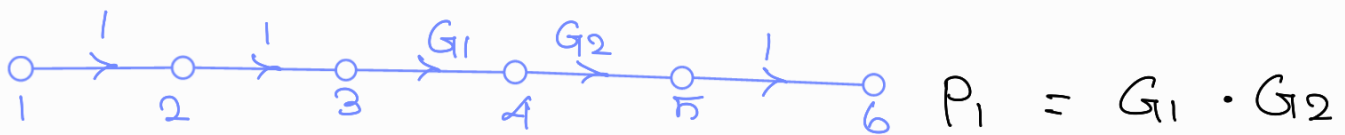
1. Convert the block diagram into signal flow graph and determine the closed loop transfer function.



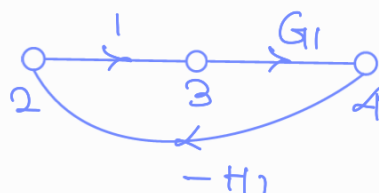
Signal flow graph:



Forward Path Gains:



Individual Loop Gains:



$$L_1 = -G_1 \cdot H_1$$

Non-touching Loops Gain:

$$N_1 = 0$$

$$\Delta = 1 - [L_1] + [0]$$

$$= 1 - [-G_1 H_1]$$

$$\Delta = 1 + G_1 H_1$$

$$\Delta_1 = 1 - 0 = 1$$

$$\Delta_2 = 1 - 0 = 1$$

Transfer Function:

$$T = \frac{\sum_{k=1}^{n=2} P_k \Delta_k}{\Delta}$$

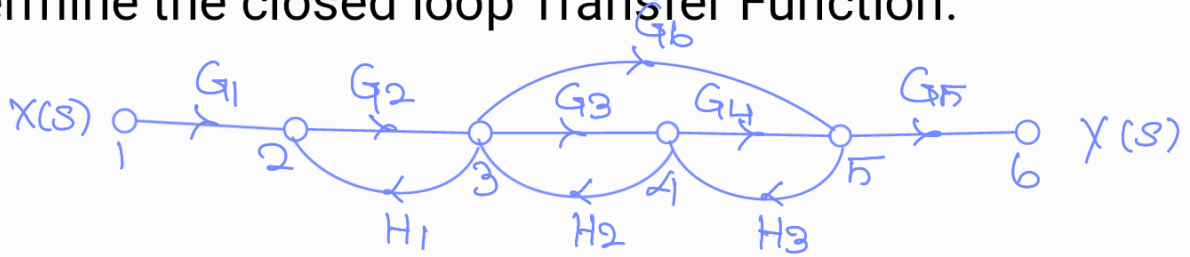
$$= \frac{[P_1 \Delta_1 + P_2 \Delta_2]}{\Delta}$$

$$= \frac{(G_1 G_2 \cdot 1) + (-G_3 \cdot 1)}{(1 + G_1 H_1)}$$

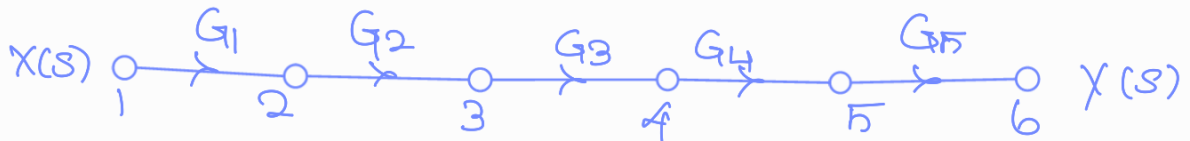
$$T = \frac{G_1 G_2 - G_3}{1 + G_1 H_1}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 G_2 - G_3}{1 + G_1 H_1}$$

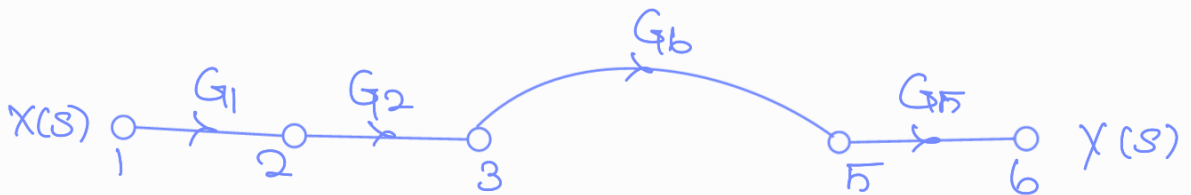
2. Determine the closed loop Transfer Function.



Forward Path Gains:

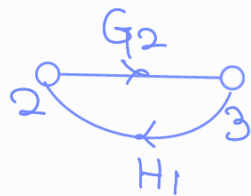


$$P_1 = G_1 \cdot G_2 \cdot G_3 \cdot G_4 \cdot G_5$$

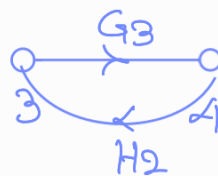


$$P_2 = G_1 \cdot G_2 \cdot G_6 \cdot G_5$$

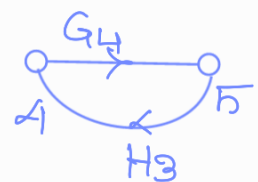
Individual Loop Gains:



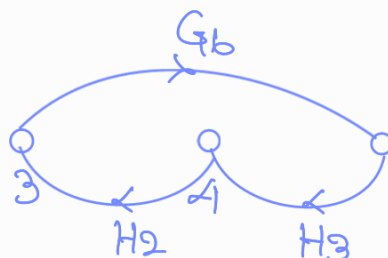
$$L_1 = G_2 H_1$$



$$L_2 = G_3 H_2$$

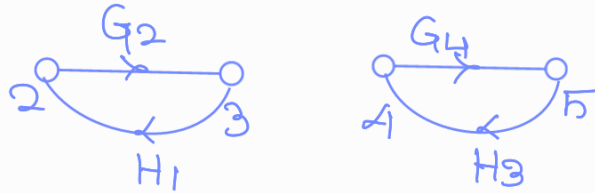


$$L_3 = G_4 H_3$$



$$L_4 = G_6 H_2 H_3$$

Two Non-touching Loops Gain:



$$N_1 = G_2 G_4 H_1 H_3$$

$$\Delta = 1 - (\sum L) + (\sum N)$$

$$= 1 - [G_2 H_1 + G_3 H_2 + G_4 H_3 + G_6 H_2 H_3] + G_2 G_4 H_1 H_3$$

$$\Delta = 1 - G_2 H_1 - G_3 H_2 - G_4 H_3 - G_6 H_2 H_3 + G_2 G_4 H_1 H_3$$

Since no part of graph is non-touching with Forward paths

$$\Delta_1 = \Delta_2 = 1$$

Transfer Function:

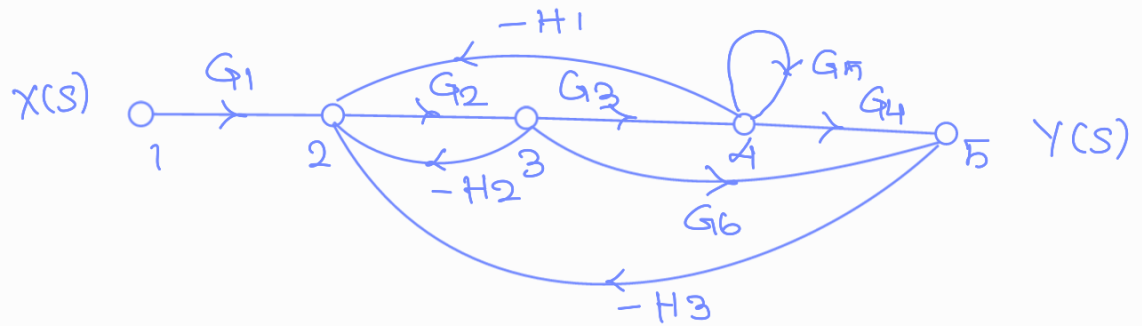
$$T = \frac{\sum_{k=1}^n P_k \Delta_k}{\Delta}$$

$$= \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

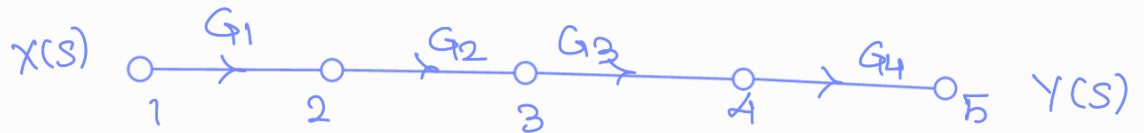
$$T = \frac{G_1 G_2 G_3 G_4 G_5 + G_1 G_2 G_5 G_6}{1 - G_2 H_1 - G_3 H_2 - G_4 H_3 - G_6 H_2 H_3 + G_2 G_4 H_1 H_3}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 G_2 G_3 G_4 G_5 + G_1 G_2 G_5 G_6}{1 - G_2 H_1 - G_3 H_2 - G_4 H_3 - G_6 H_2 H_3 + G_2 G_4 H_1 H_3}$$

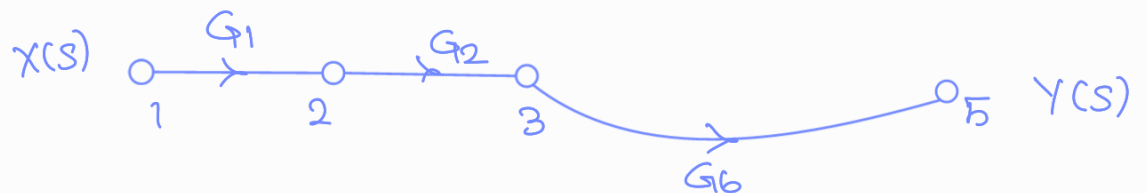
3. Find the overall gain.



Forward Path Gains:

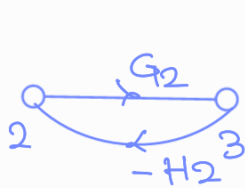


$$P_1 = G_1 \cdot G_2 \cdot G_3 \cdot G_4$$



$$P_2 = G_1 \cdot G_2 \cdot G_6$$

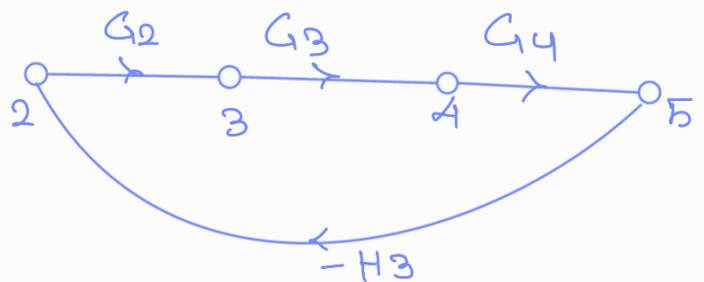
Individual Loop Gains:



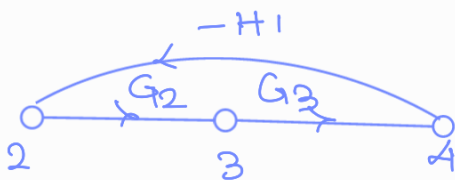
$$L_1 = -G_2 H_2$$



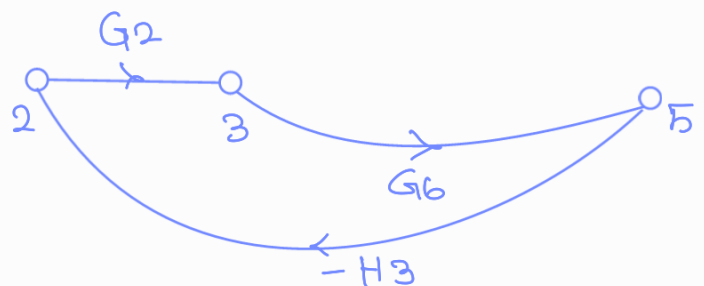
$$L_2 = G_5$$



$$L_3 = -G_2 G_3 G_4 H_3$$

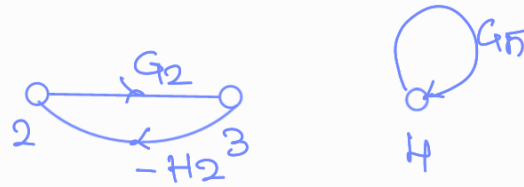


$$L_4 = -G_2 G_3 H_1$$

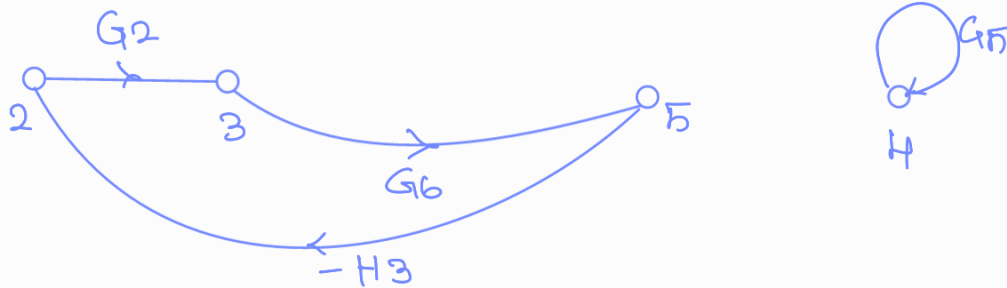


$$L_5 = -G_2 G_6 H_3$$

Two Non-touching Loops Gain:



$$N_1 = -G_2 G_5 H_2$$



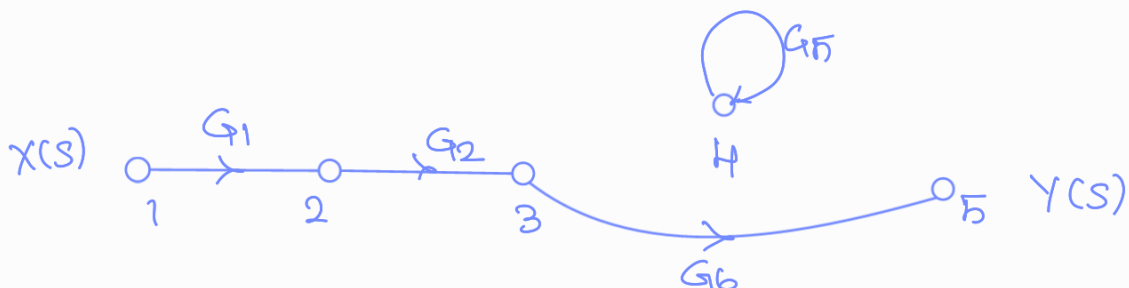
$$N_2 = -G_2 G_5 G_6 H_3$$

$$\Delta = 1 - (\sum L) + (\sum N)$$

$$= 1 - (-G_2 H_2 + G_5 - G_2 G_3 G_4 H_3 - G_2 G_3 H_1 - G_2 G_6 H_3) + (-G_2 G_5 H_2 - G_2 G_5 G_6 H_3)$$

$$\Delta = 1 + G_2 H_2 - G_5 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$

$$\Delta_1 = 1$$



$$\Delta_2 = 1 - G_5$$

Transfer Function:

$$T = \frac{\sum_{k=1}^n P_k A_k}{\Delta}$$

$$= \frac{P_1 A_1 + P_2 A_2}{\Delta}$$

$$= \frac{(G_1 G_2 G_3 G_4 \cdot 1) + [G_1 G_2 G_6 \cdot (1 - G_5)]}{\Delta}$$

$$\Delta = 1 + G_2 H_2 - G_5 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$

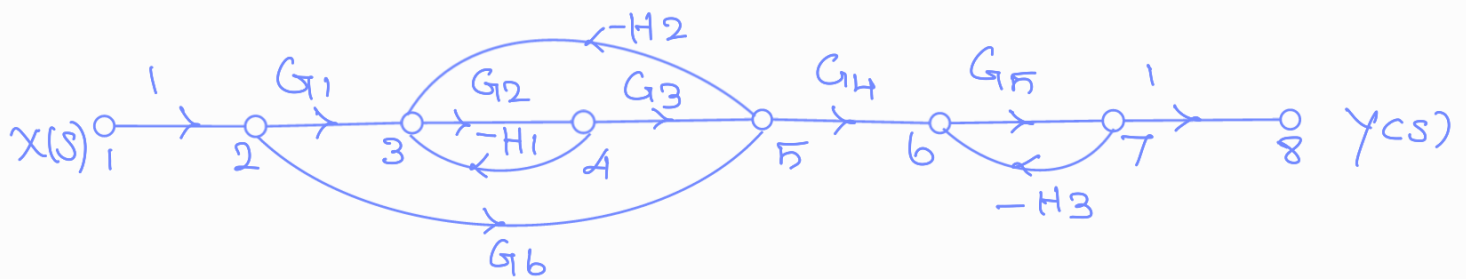
$$T = \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_6 - G_1 G_2 G_5 G_6}{\Delta}$$

$$\Delta = 1 + G_2 H_2 - G_5 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$

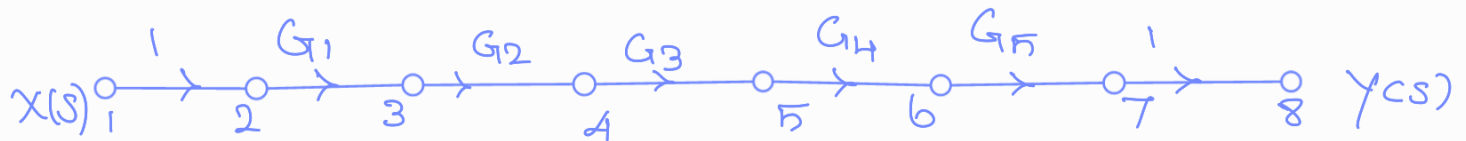
$$\frac{Y(s)}{X(s)} = \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_6 - G_1 G_2 G_5 G_6}{\Delta}$$

$$\Delta = 1 + G_2 H_2 - G_5 + G_2 G_3 G_4 H_3 + G_2 G_3 H_1 + G_2 G_6 H_3 - G_2 G_5 H_2 - G_2 G_5 G_6 H_3$$

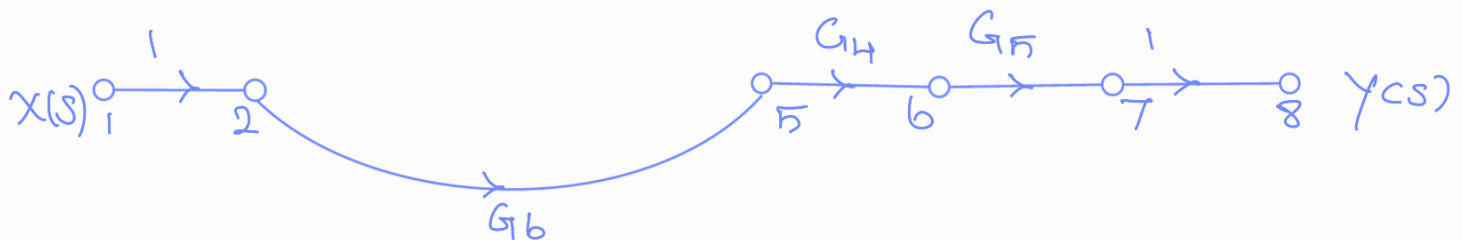
4. Find overall transfer function of the system.



Forward Path Gain:



$$P_1 = G_1 \cdot G_2 \cdot G_3 \cdot G_4 \cdot G_5$$

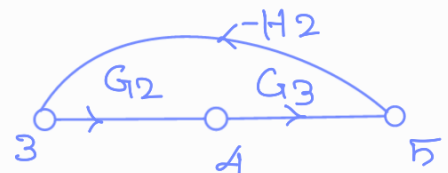


$$P_2 = G_4 \cdot G_5 \cdot G_6$$

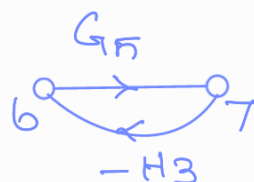
Individual Loop Gains:



$$L_1 = -G_2 H_1$$



$$L_2 = -G_2 G_3 H_2$$

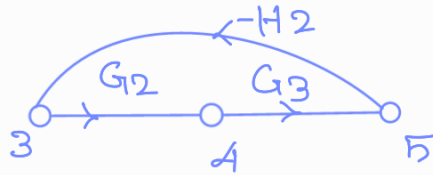
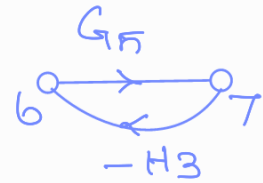


$$L_3 = -G_5 H_3$$

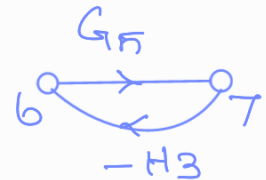
Two Non-touching Loops Gain:



$$N_1 = G_2 G_5 H_1 H_3$$



$$N_2 = G_2 G_3 G_5 H_2 H_3$$

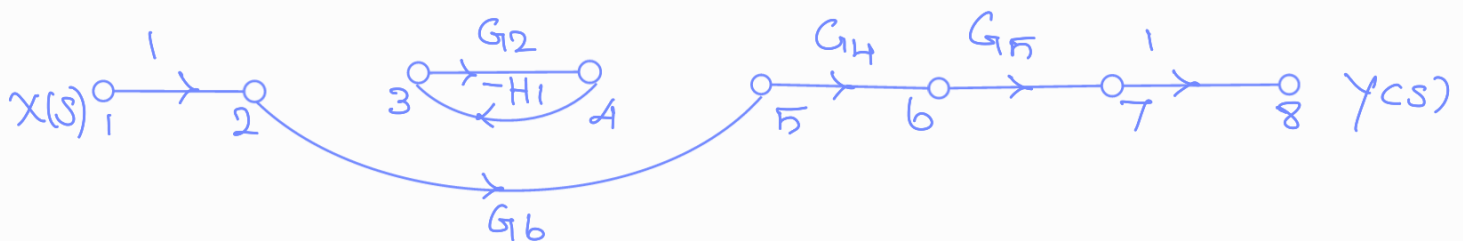


$$\Delta = 1 - \sum L + \sum N$$

$$= 1 - (-G_2 H_1 - G_2 G_3 H_2 - G_5 H_3) + (G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3)$$

$$\Delta = 1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3$$

$$\Delta_1 = 1$$



$$\Delta_2 = 1 - (-G_2 H_1)$$

$$\Delta_2 = 1 + G_2 H_1$$

Transfer Function:

$$T = \frac{\sum_{k=1}^{k=2} P_k \Delta_k}{\Delta}$$

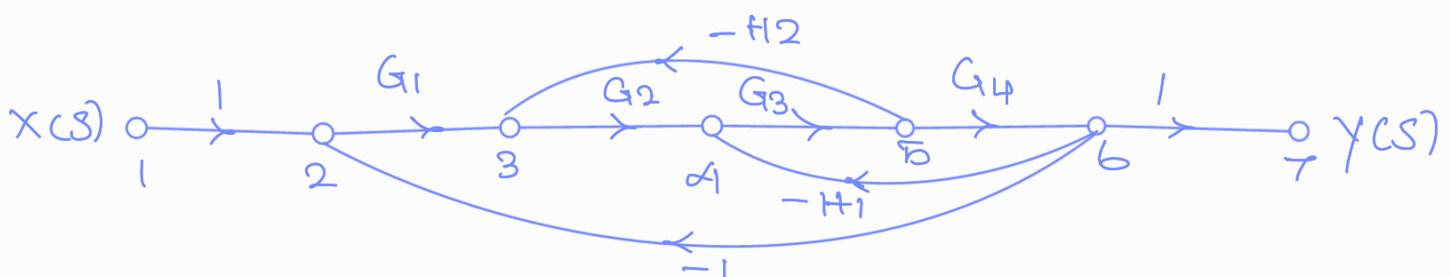
$$= \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

$$= \frac{G_1 G_2 G_3 G_4 G_5 + G_4 G_5 G_6 (1 + G_2 H_1)}{1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3}$$

$$T = \frac{G_1 G_2 G_3 G_4 G_5 + G_4 G_5 G_6 + G_2 G_4 G_5 G_6 H_1}{1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 G_2 G_3 G_4 G_5 + G_4 G_5 G_6 + G_2 G_4 G_5 G_6 H_1}{1 + G_2 H_1 + G_2 G_3 H_2 + G_5 H_3 + G_2 G_5 H_1 H_3 + G_2 G_3 G_5 H_2 H_3}$$

5. Find the overall gain of the system.

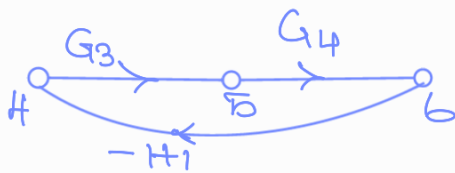


Forward Path Gains:

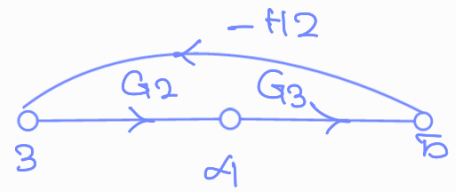


$$P_1 = G_1 \cdot G_2 \cdot G_3 \cdot G_4$$

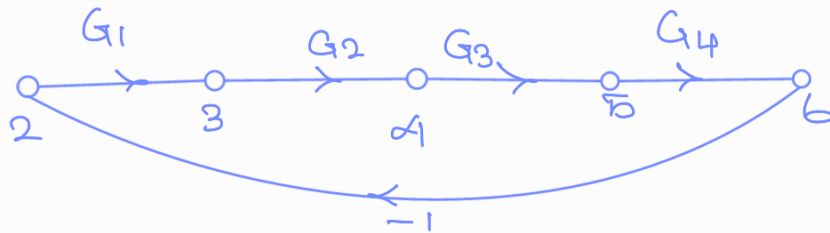
Individual Loop Gains:



$$L_1 = -G_3 G_4 H_1$$



$$L_2 = -G_2 G_3 H_2$$



$$L_3 = -G_1 G_2 G_3 G_4$$

Two Non-touching Loops Gain:

$$N_c = 0$$

$$\Delta = 1 - \sum L + \sum N$$

$$= 1 - [-G_3 G_4 H_1 - G_2 G_3 H_2 - G_1 G_2 G_3 G_4]$$

$$\Delta = 1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4$$

$$\Delta_1 = 1 - 0 = 1$$

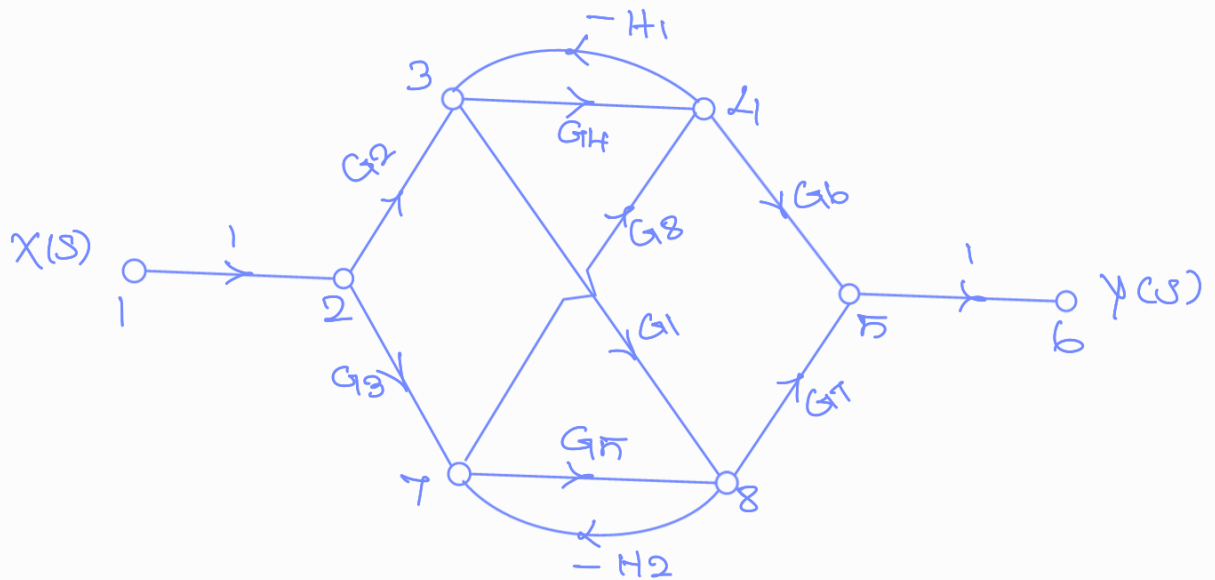
Transfer Function:

$$T = \frac{P_1 \Delta_1}{\Delta}$$

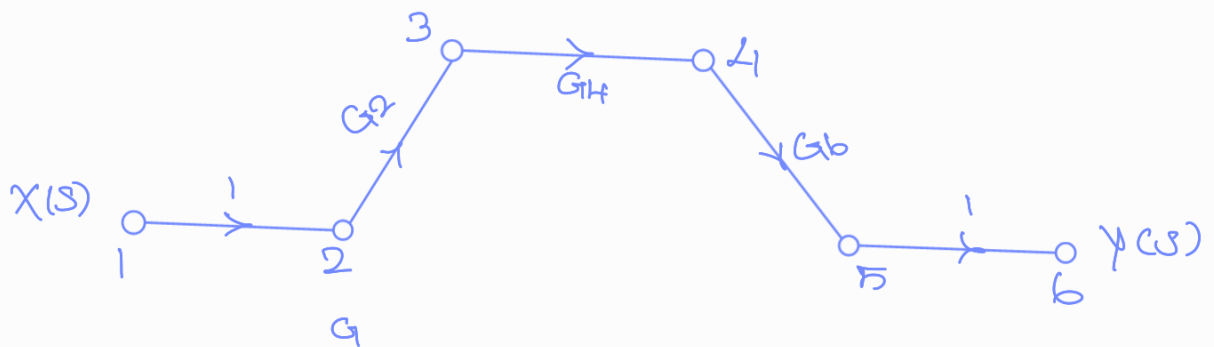
$$T = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4}$$

$$\frac{Y(s)}{X(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4}$$

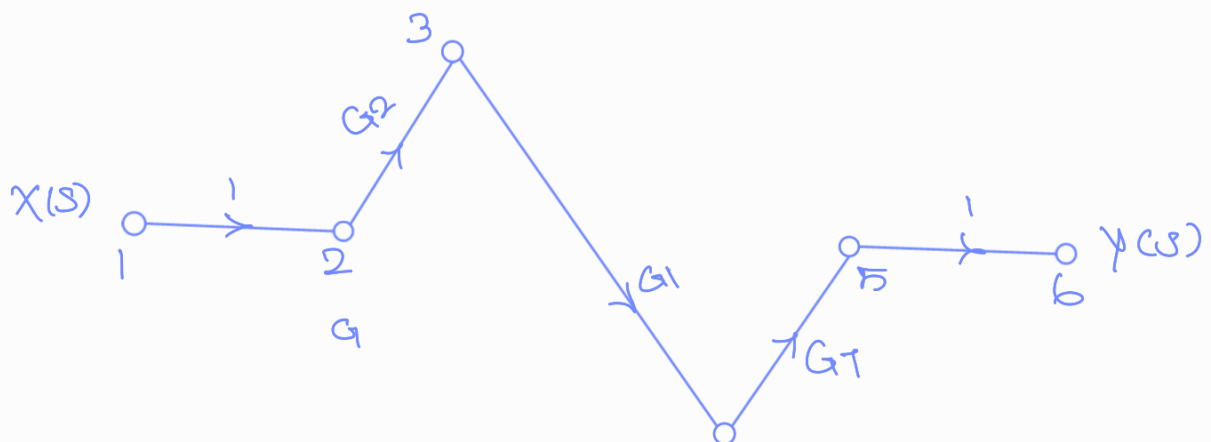
6. Determine the transfer Function using Mason's gain formula.



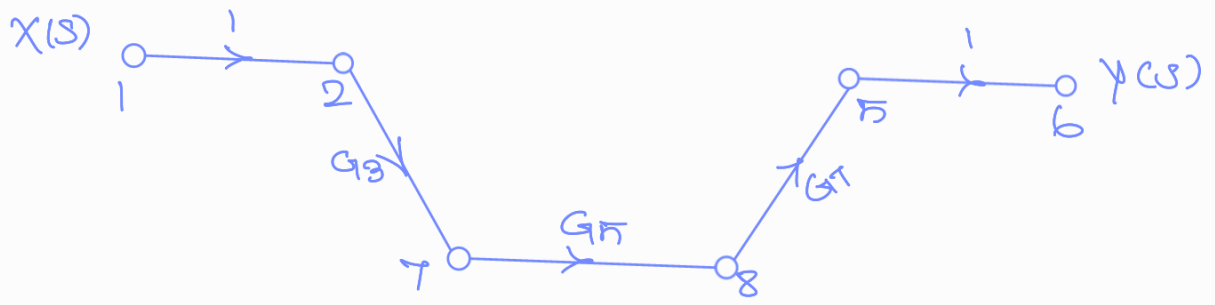
Forward Path Gains:



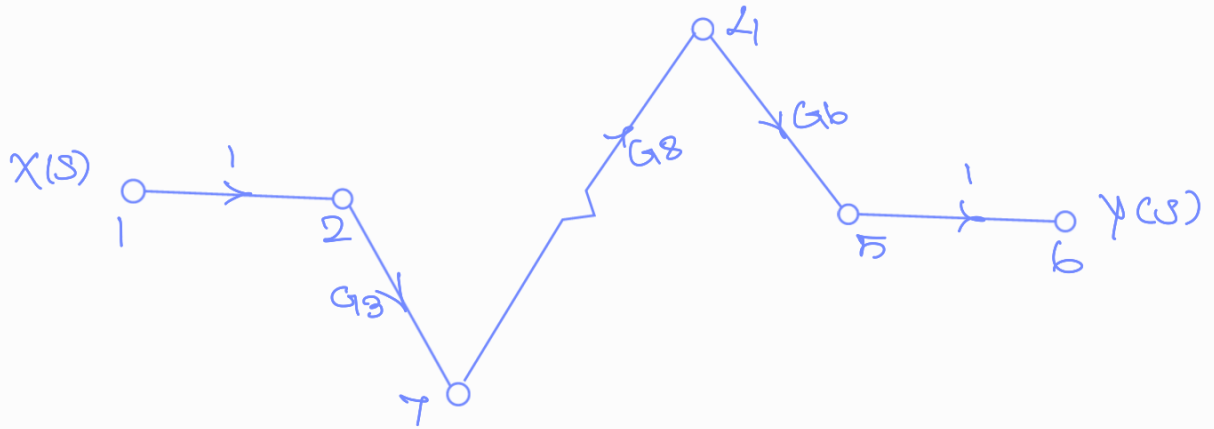
$$P_1 = G_2 G_4 G_6$$



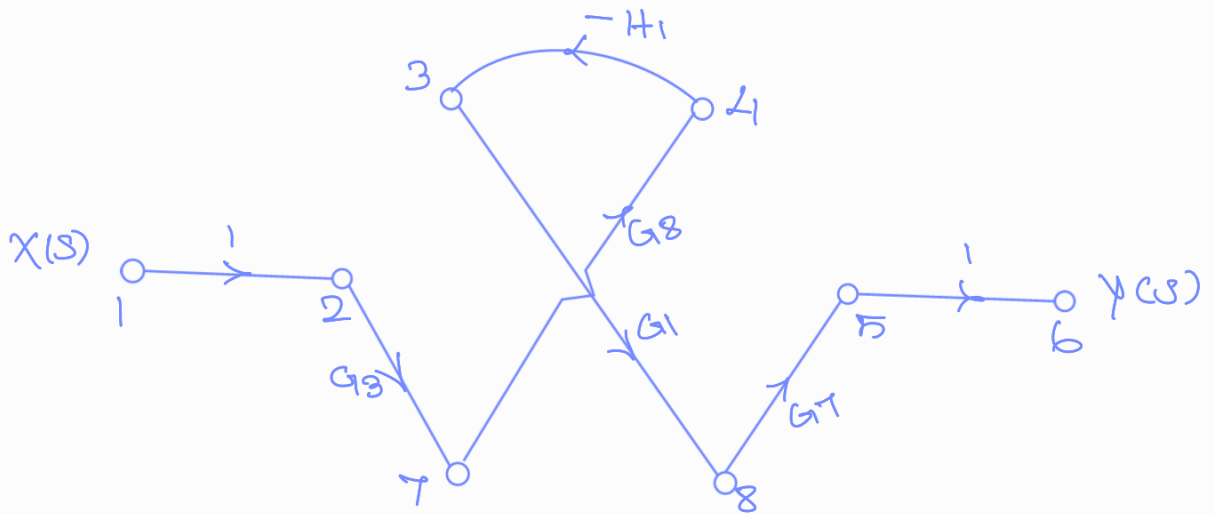
$$P_2 = G_3 G_5$$



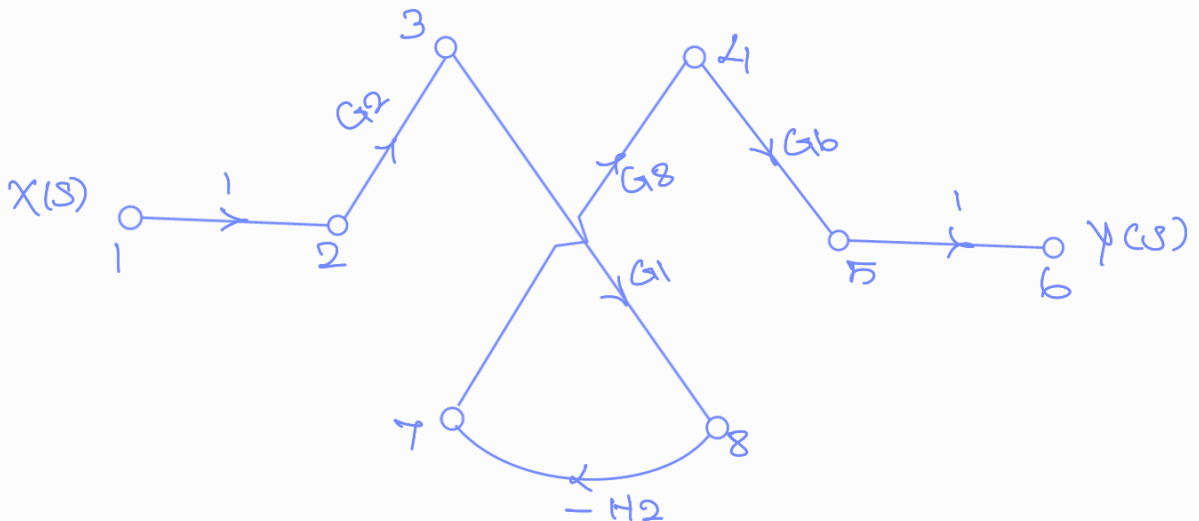
$$P_3 = G_3 \cdot G_5 \cdot G_4$$



$$P_4 = G_3 \cdot G_6 \cdot G_8$$

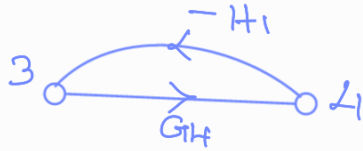


$$P_5 = -G_1 G_3 G_4 G_8 H_1$$

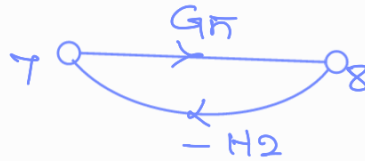


$$P_6 = -G_1 G_2 G_4 G_8 H_2$$

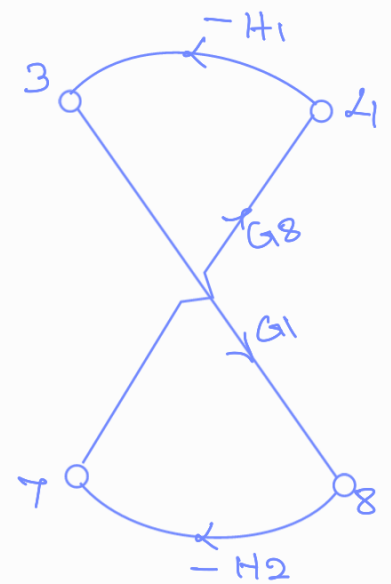
Individual Loop Gains:



$$L_1 = -G_4 H_1$$

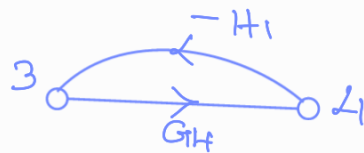


$$L_2 = -G_5 H_2$$



$$L_3 = G_1 G_8 H_1 H_2$$

Two Non-touching Loops Gain:



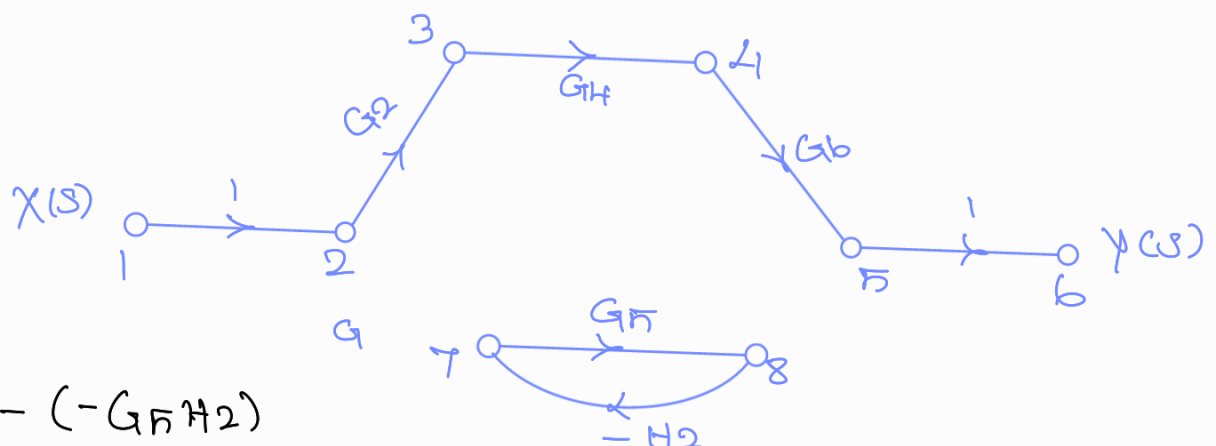
$$N_1 = G_4 G_5 H_1 H_2$$

$$\Delta = 1 - \sum L + \sum N$$

$$\Delta = 1 - [L_1 + L_2 + L_3] + N_1$$

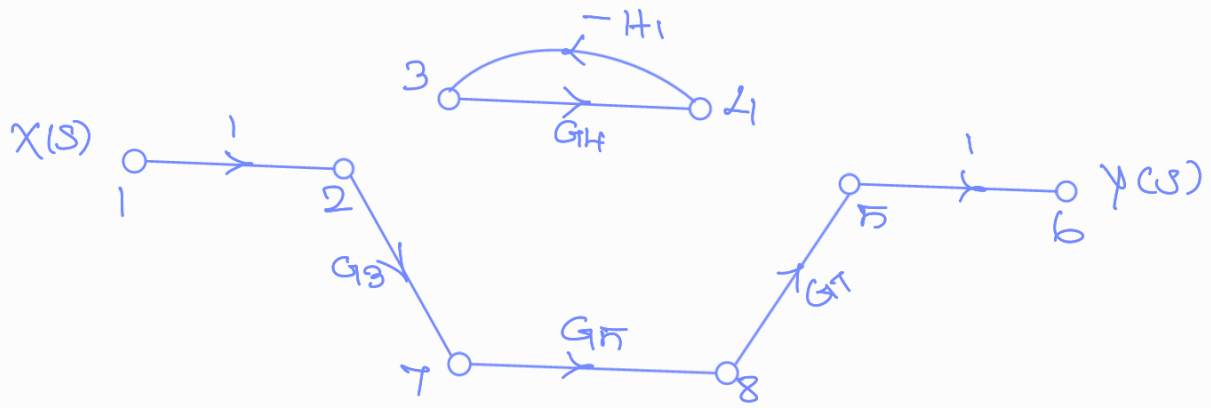
$$= 1 - [-G_4 H_1 - G_5 H_2 + G_1 G_8 H_1 H_2] + G_4 G_5 H_1 H_2$$

$$\Delta = 1 + G_4 H_1 + G_5 H_2 - G_1 G_8 H_1 H_2 + G_4 G_5 H_1 H_2$$



$$\Delta_1 = 1 - (-G_5 H_2)$$

$$\Delta_1 = 1 + G_5 H_2$$



$$\Delta_3 = 1 - (-G_4 H_1)$$

$$\Delta_3 = 1 + G_4 H_1$$

$$\Delta_2 = \Delta_4 = \Delta_5 = \Delta_6 = 1$$

Transfer Function:

$$T = \frac{\sum_{k=1}^{n=6} P_k \Delta_k}{\Delta}$$

$$= \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3 + P_4 \Delta_4 + P_5 \Delta_5 + P_6 \Delta_6]$$

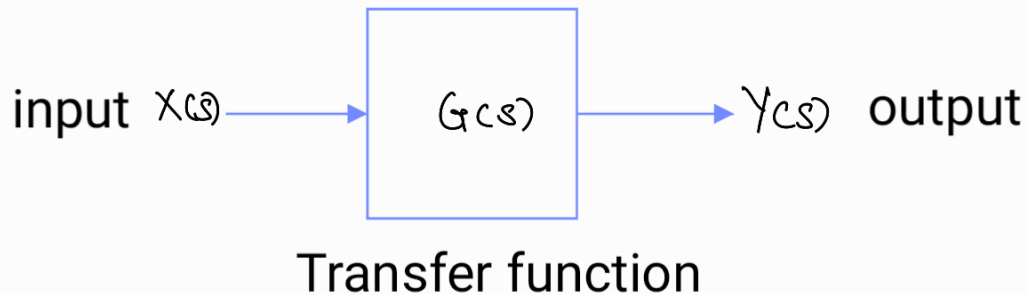
$$T = \frac{G_2 G_4 G_6 (1 + G_5 H_2) + G_1 G_2 G_7 + G_3 G_5 G_7 (1 + G_4 H_1) + G_3 G_6 G_8 - G_1 G_3 G_7 G_8 H_1 - G_1 G_2 G_6 G_8 H_2}{1 + G_4 H_1 + G_5 H_2 - G_1 G_8 H_1 H_2 + G_4 G_5 H_1 H_2}$$

Block diagram

→ pictorial representation of the functions performed by each component and of the flow of signals in a system.

→ elements of block diagram are block, branch point and the summing point.

→ block: a symbol for mathematical operations on the input signals to the block that produces the output.



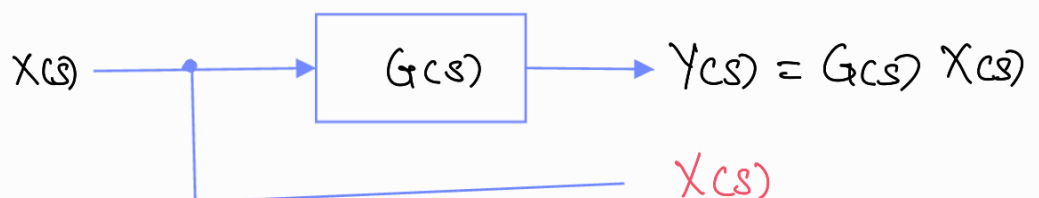
↳ input: arrowhead toward the block.

↳ output: arrowhead leaving away from the block.

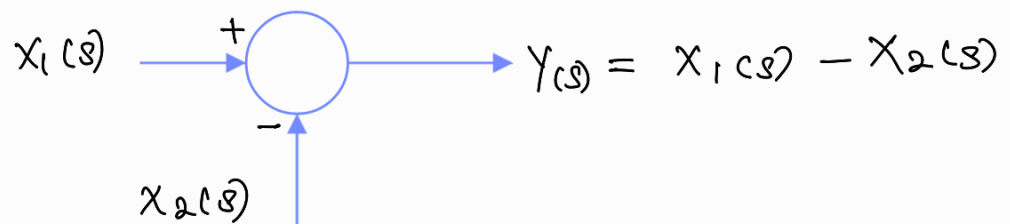
↳ output of the block is given by the product of input signal and the block transfer function.

$$Y(s) = G(s) X(s)$$

→ branch point: a point from which the signals from block goes concurrently to other blocks or summing points.



→ summing point: used to add/subtract two or more signals in the system.



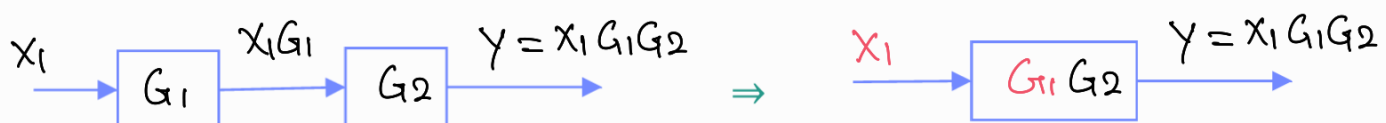
→ Governing equation gives overall block diagram of the system.

Block diagram reduction

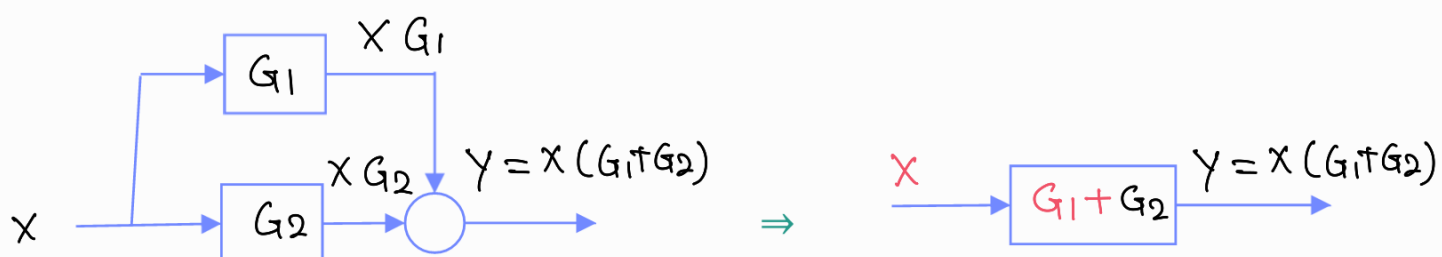
→ block diagram can be reduced to find the overall gain of the system.

→ rules are framed such that any modification made on the diagram does not alter the input - output relation.

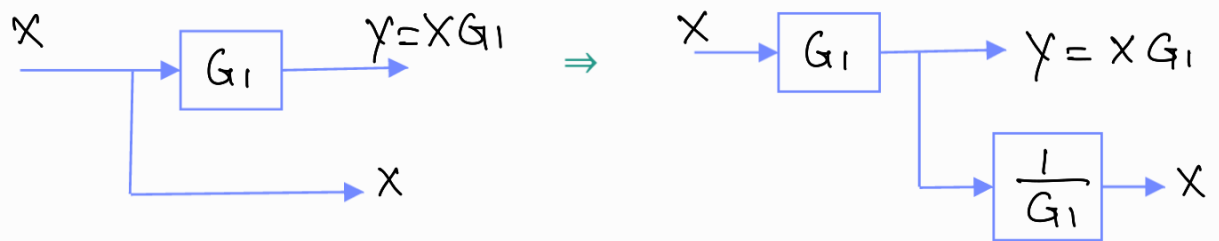
↳ combining the blocks in cascade:



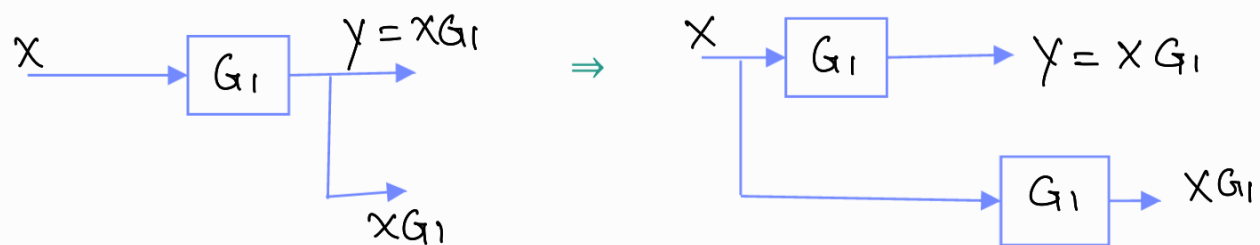
↳ combining the parallel blocks or feed forward paths:



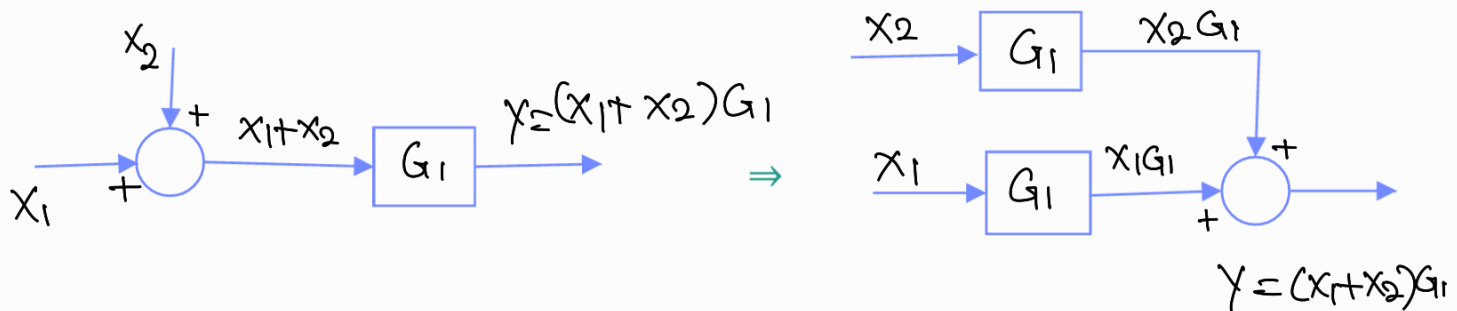
↳ moving the branch point ahead of the block:



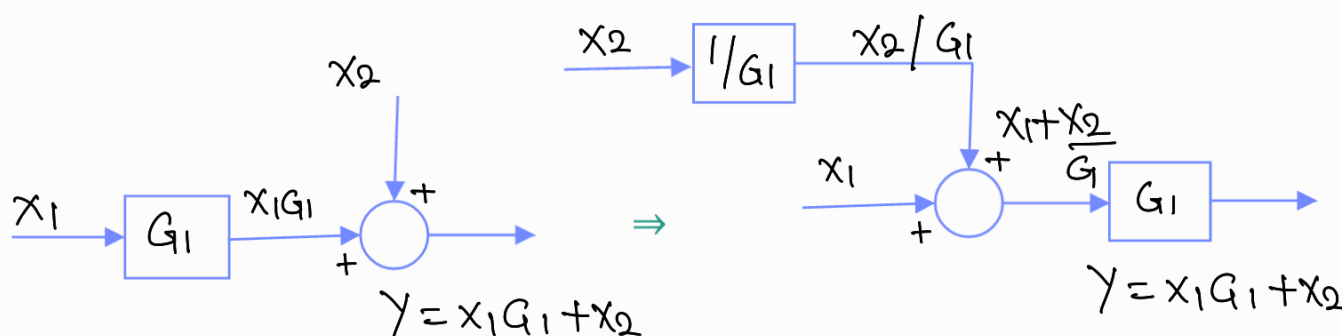
↳ moving the branch point before the block:



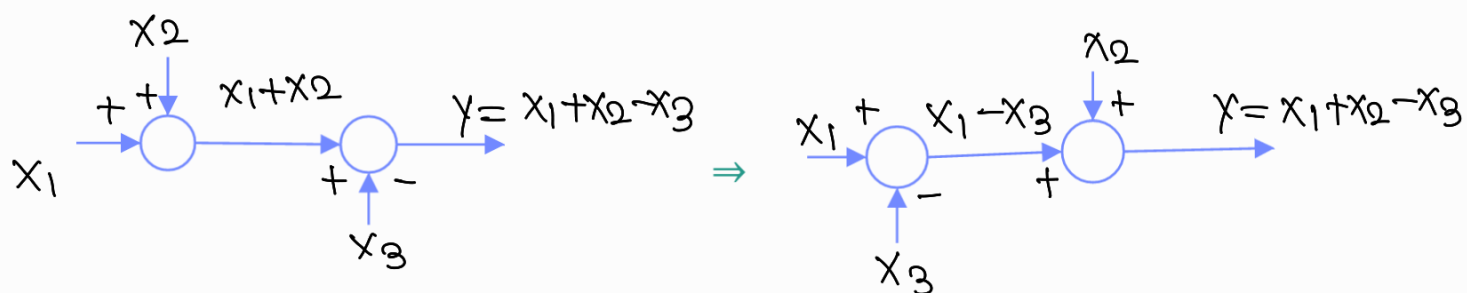
↳ moving the summing point ahead of the block:



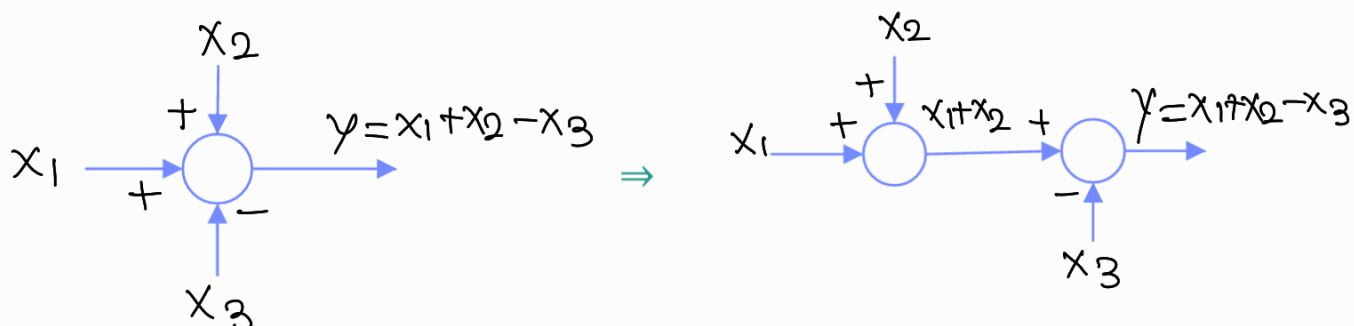
↳ moving the summing point before the block:



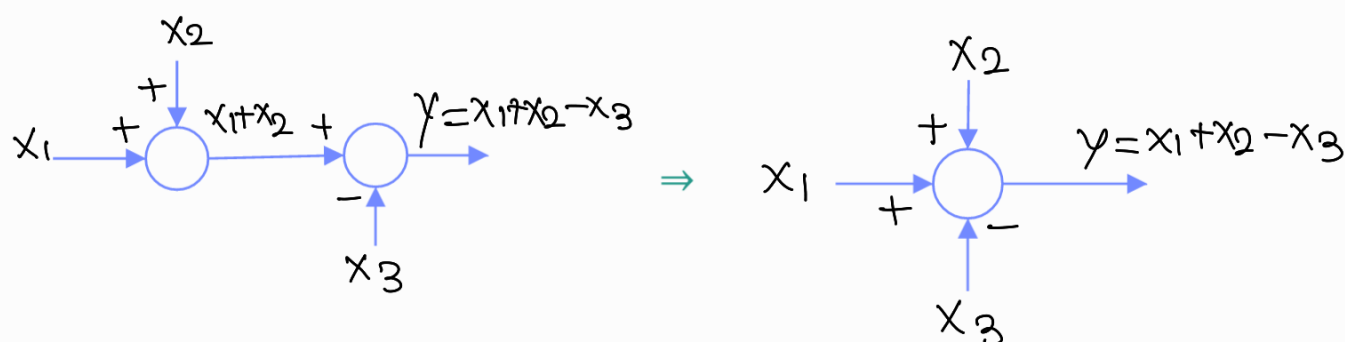
↳ interchanging summing points:



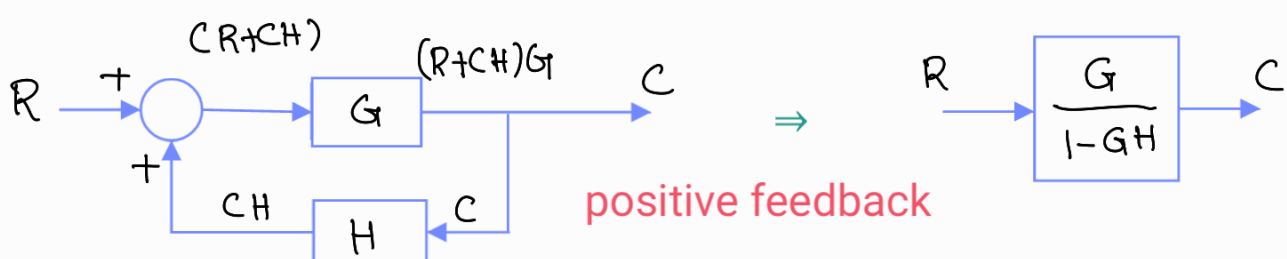
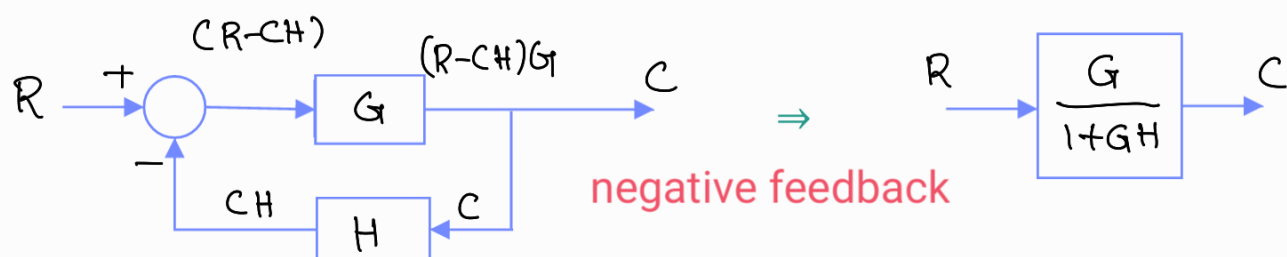
↳ splitting summing points:



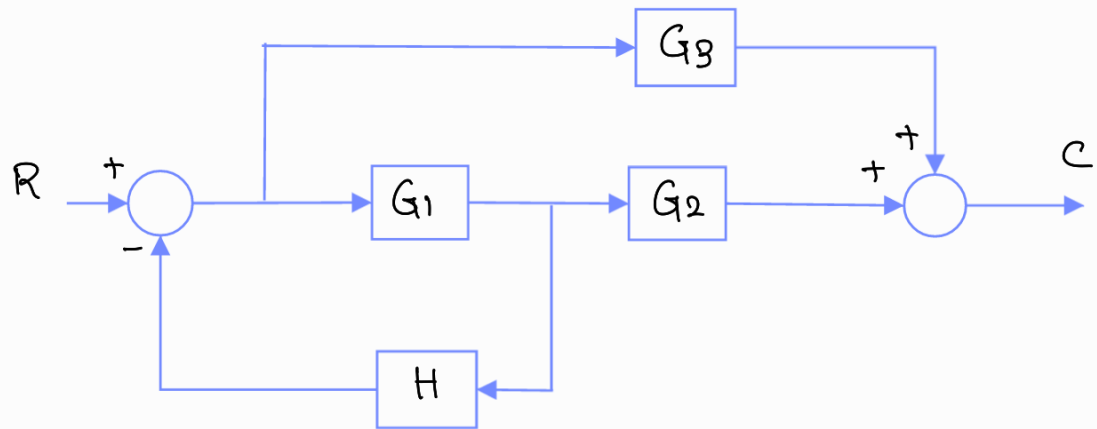
↳ combining summing points:



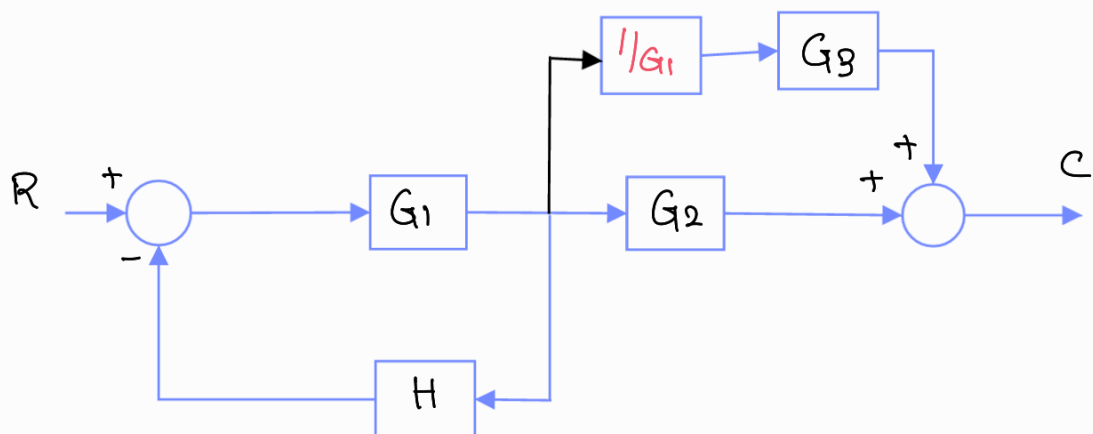
↳ elimination of feedback loop:



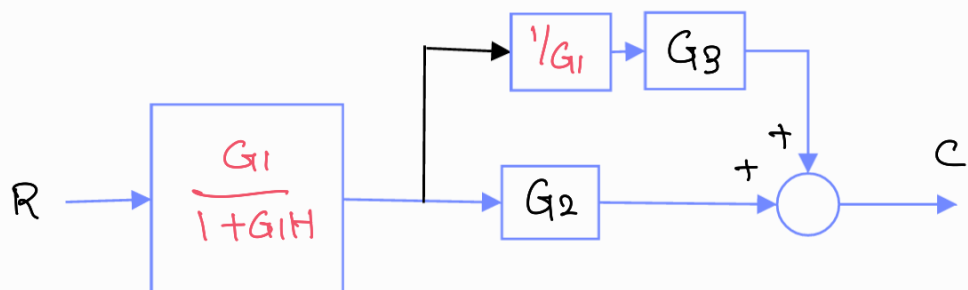
1. Reduce the block diagram and find (C / R).



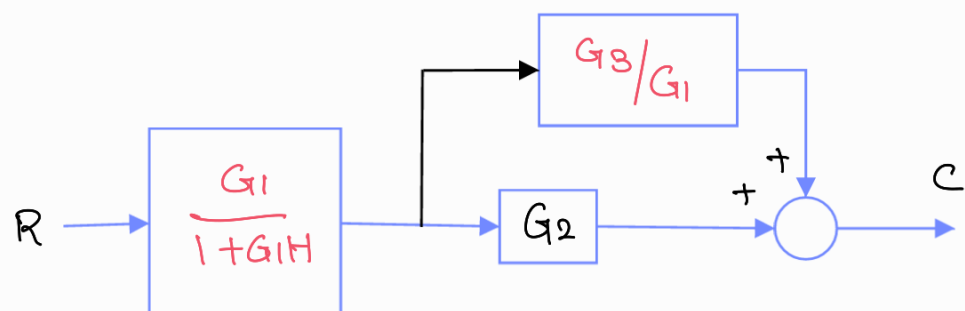
→ move the branch point after the block:



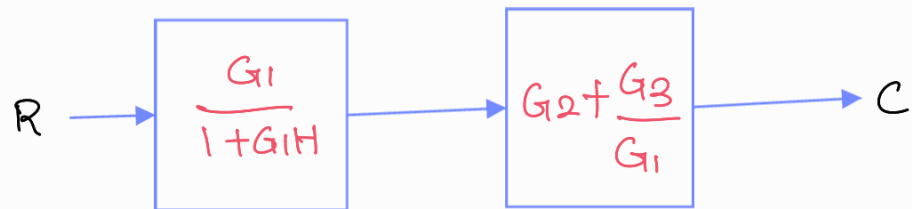
→ eliminate the feedback path:



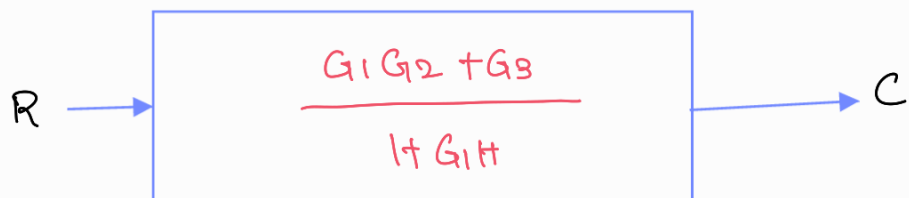
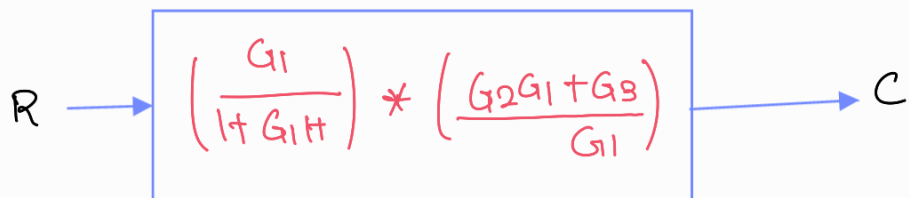
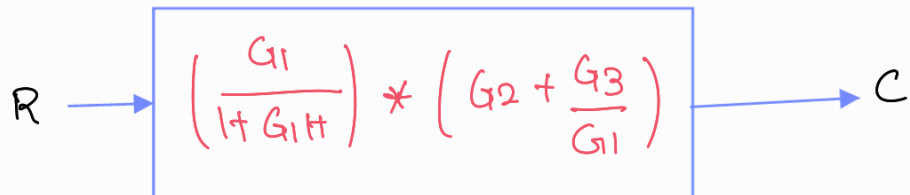
→ combining blocks in cascade:



→ combining parallel blocks:



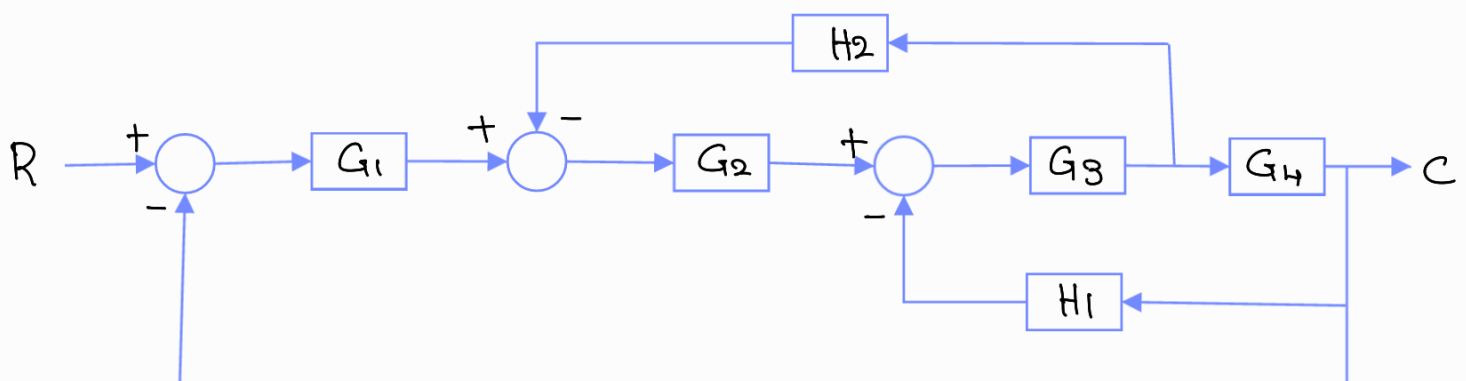
→ combining blocks in cascade:



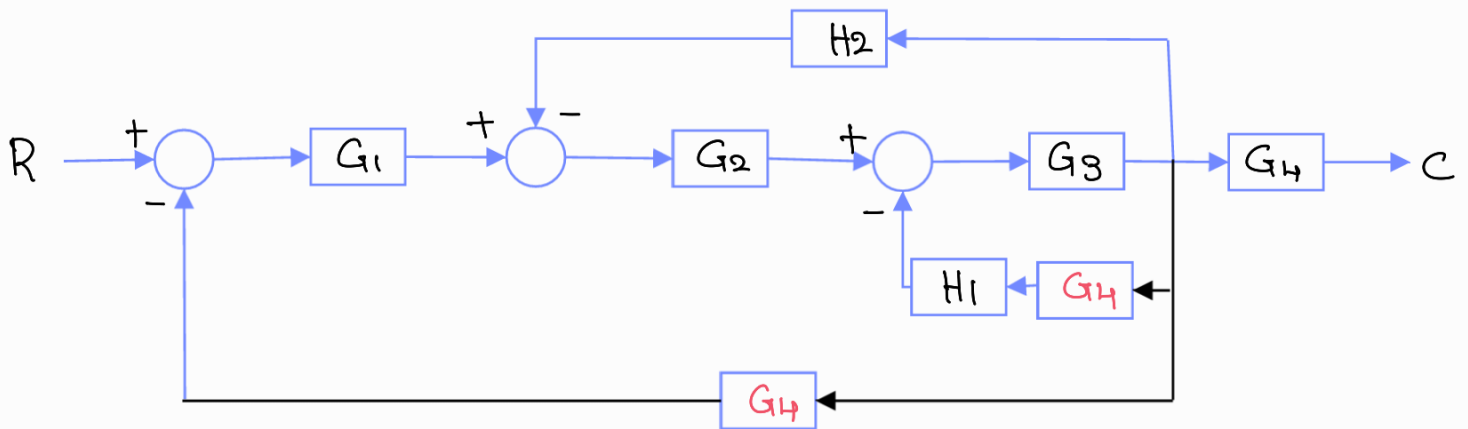
→ overall transfer function:

$$\frac{C}{R} = \frac{G_1G_2 + G_3}{1+G_1H}$$

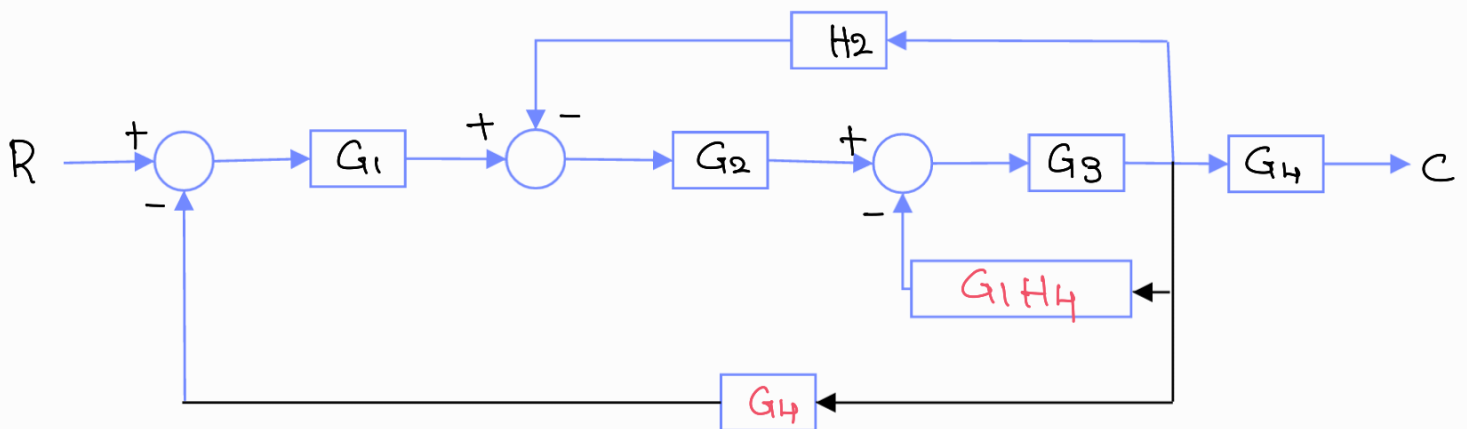
2. Determine the overall transfer function for the given system.



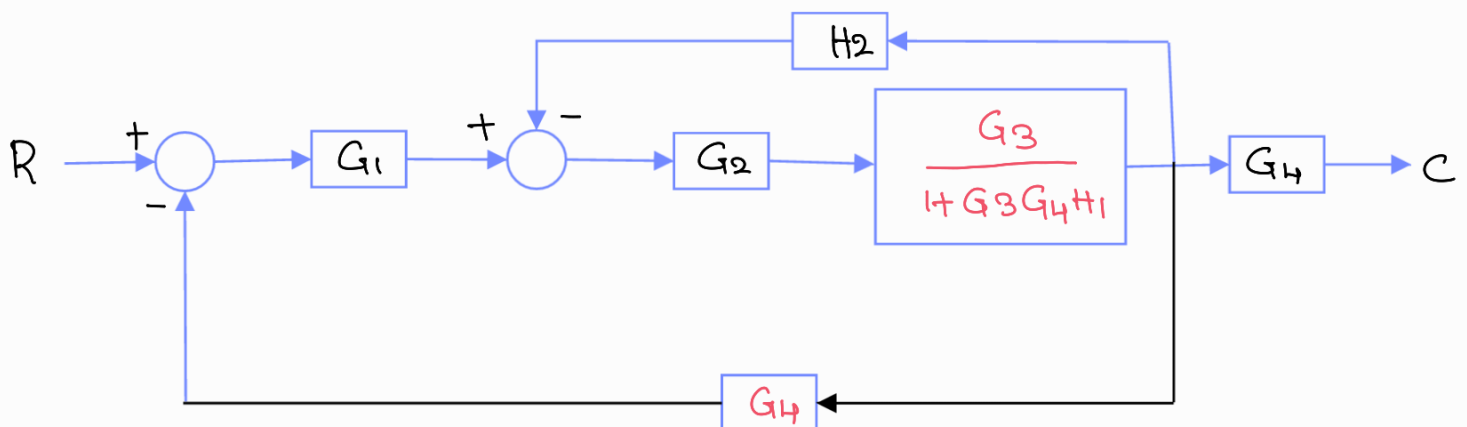
→ moving the branch point before the block:



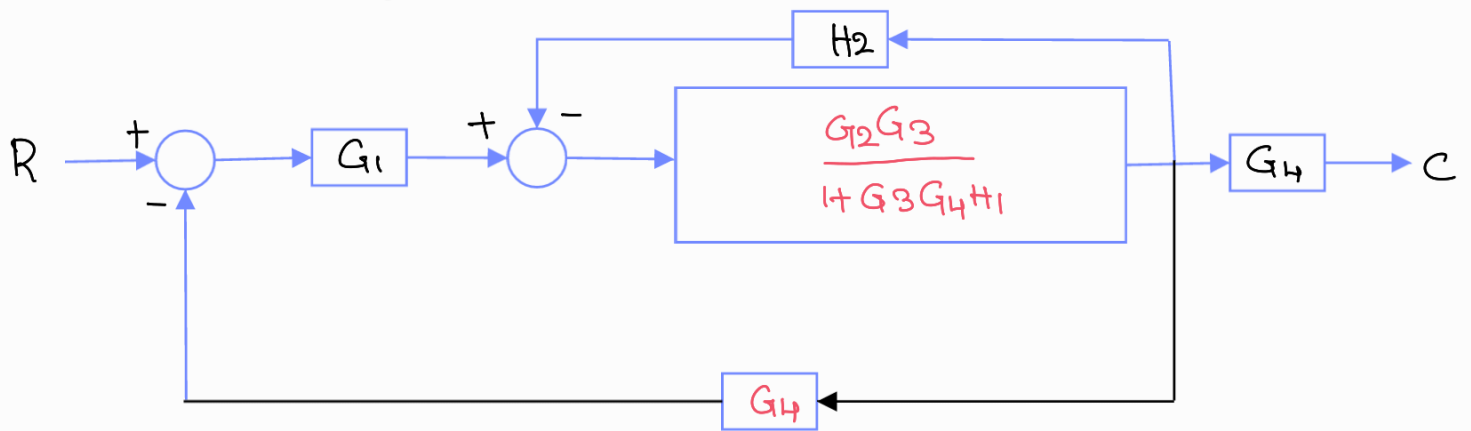
→ combining the blocks in cascade:



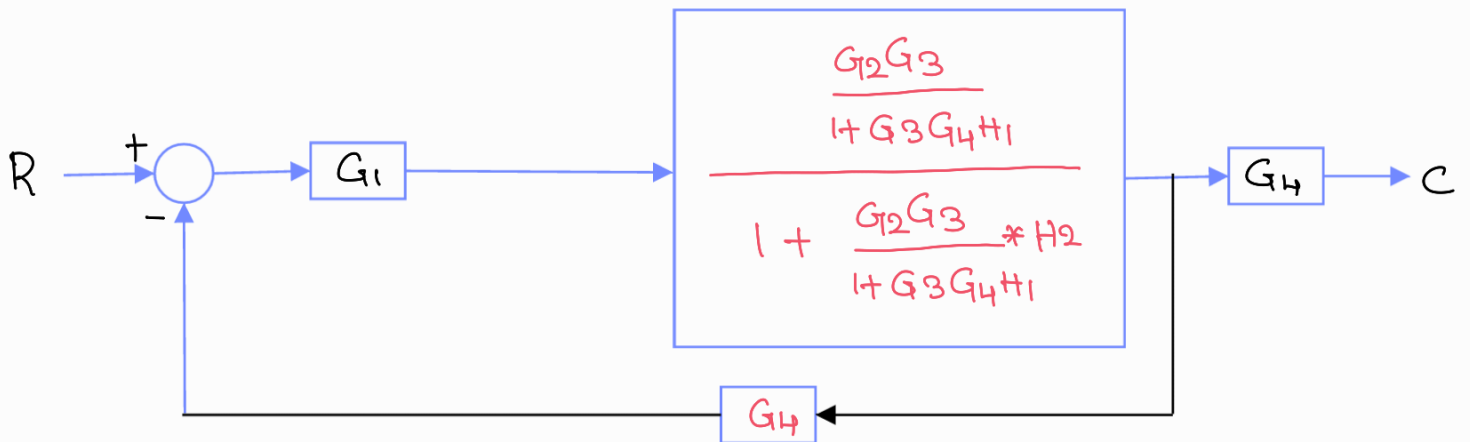
→ eliminate the feedback path:



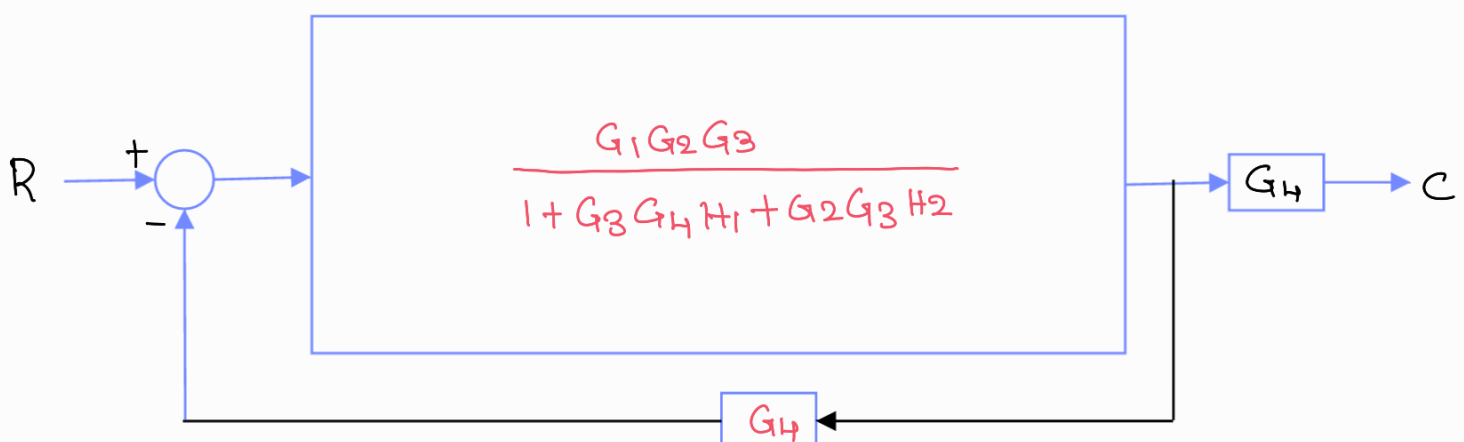
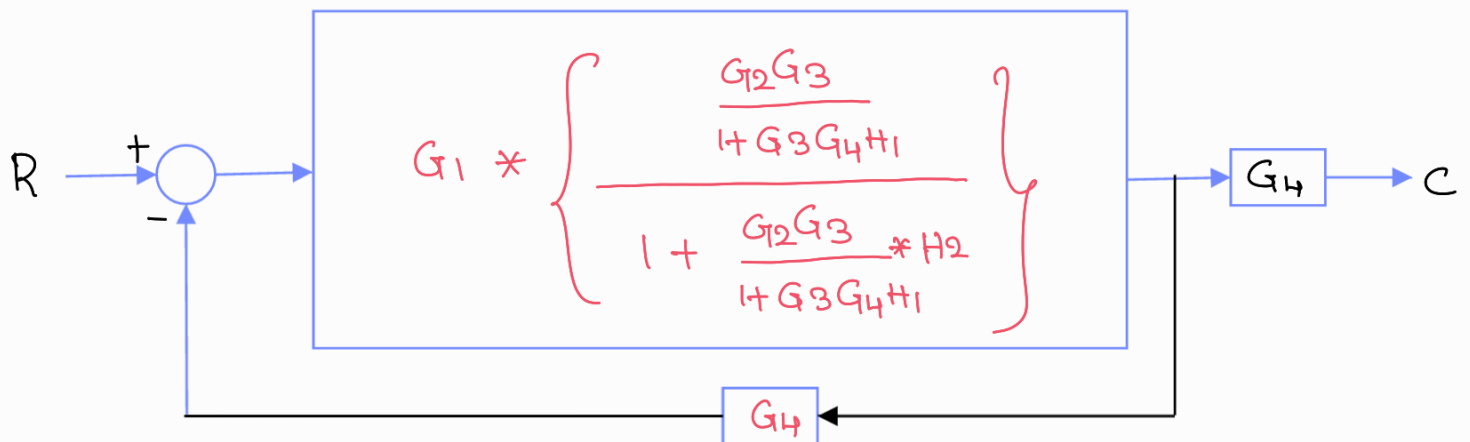
→ combining the blocks in cascade:



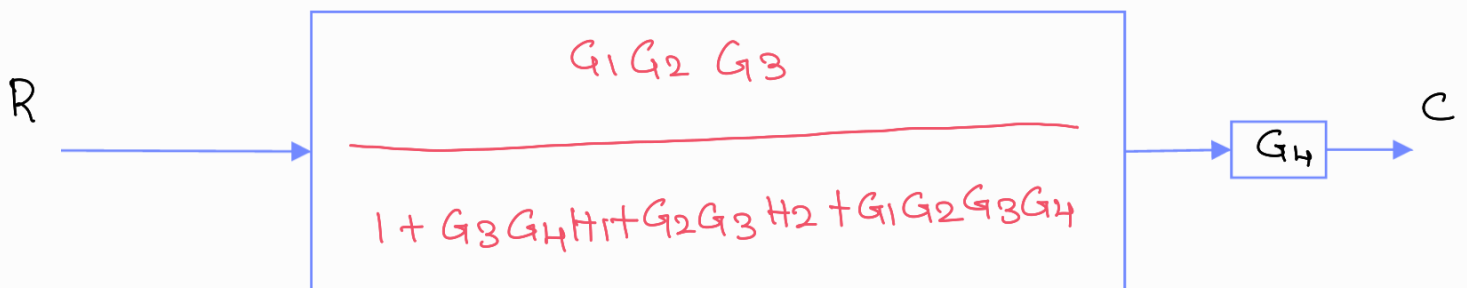
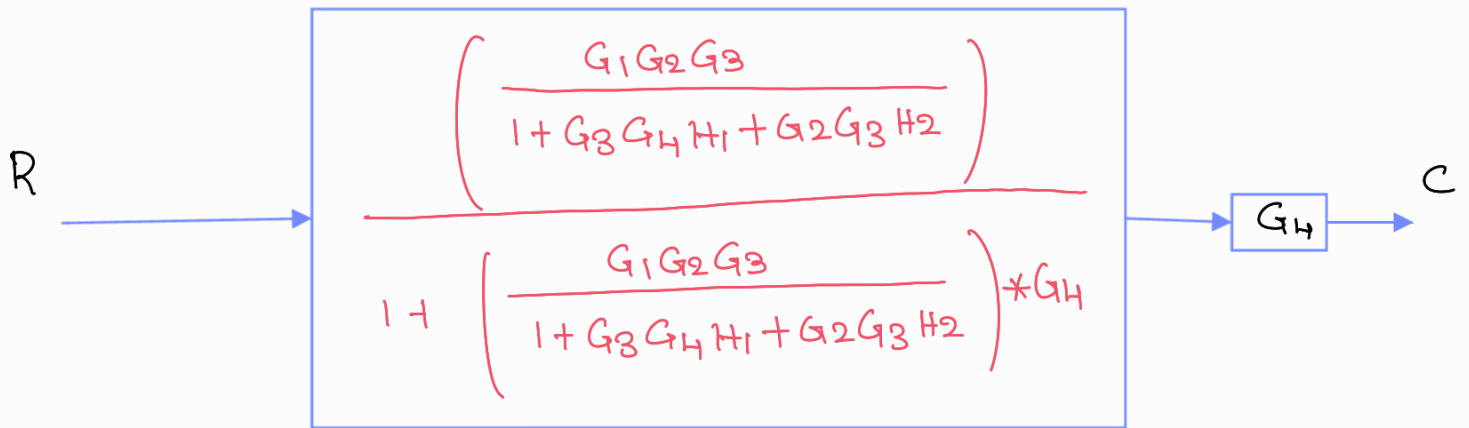
→ eliminate the feedback path:



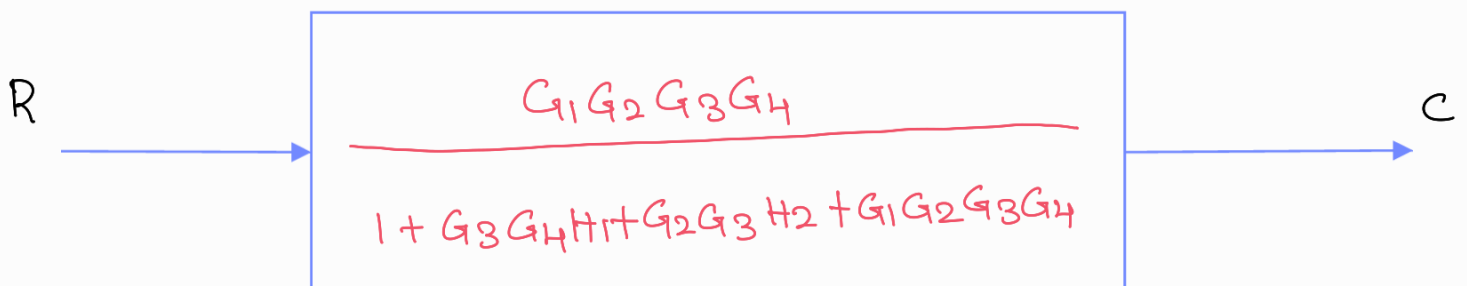
→ combining the blocks in cascade:



→ eliminate the feedback path:



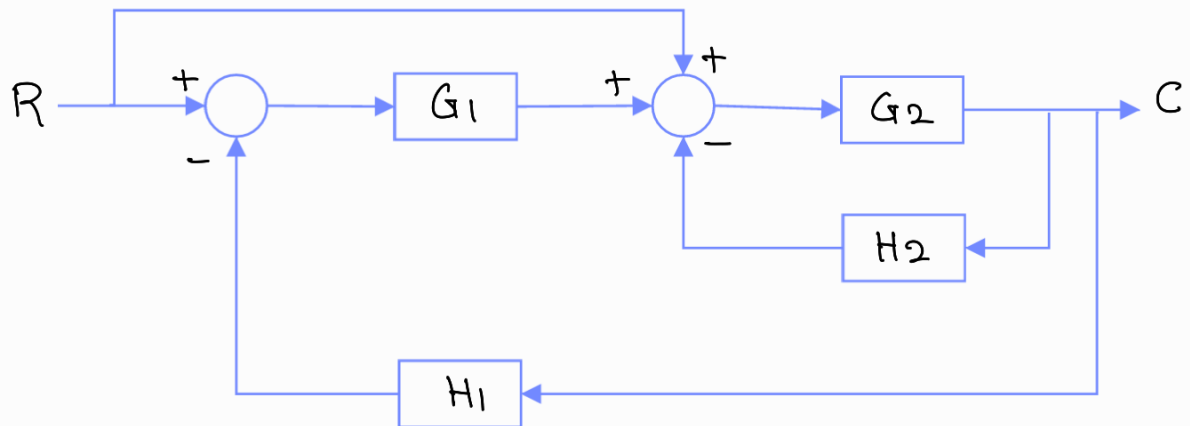
→ combining the blocks in cascade:



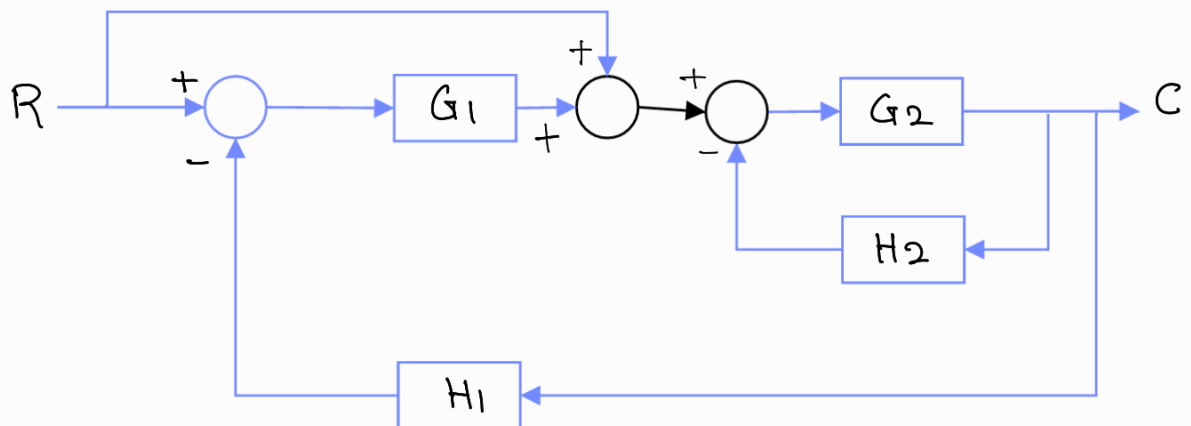
→ overall transfer function:

$$\frac{C}{R} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4}$$

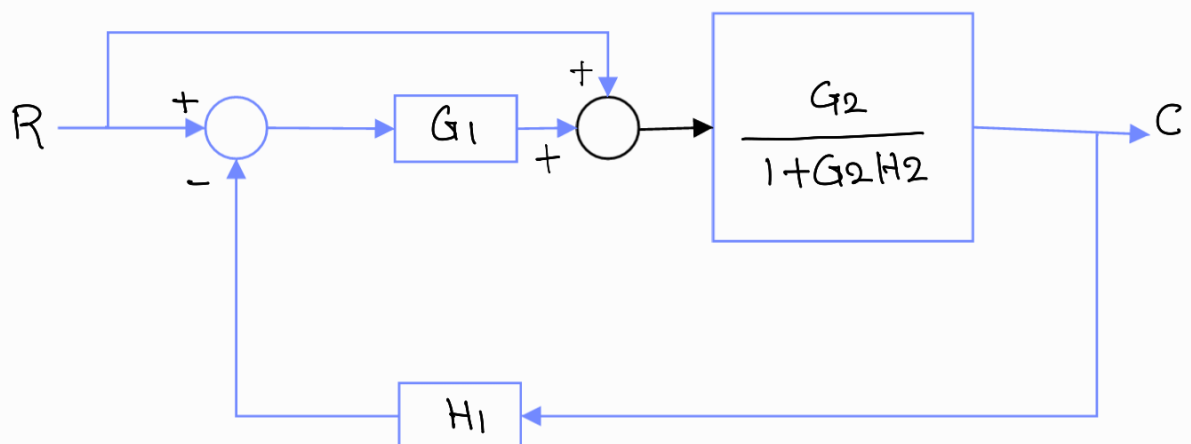
3. Using the block diagram technique, determine the closed loop transfer function.



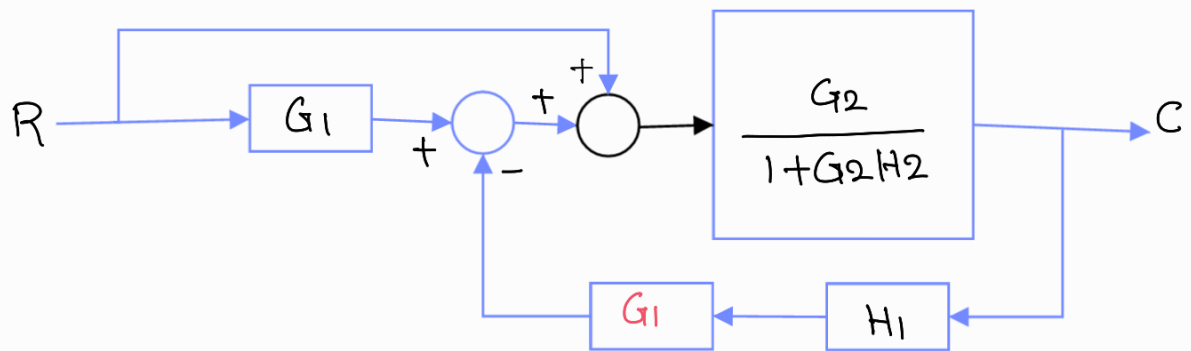
→ splitting the summing points:



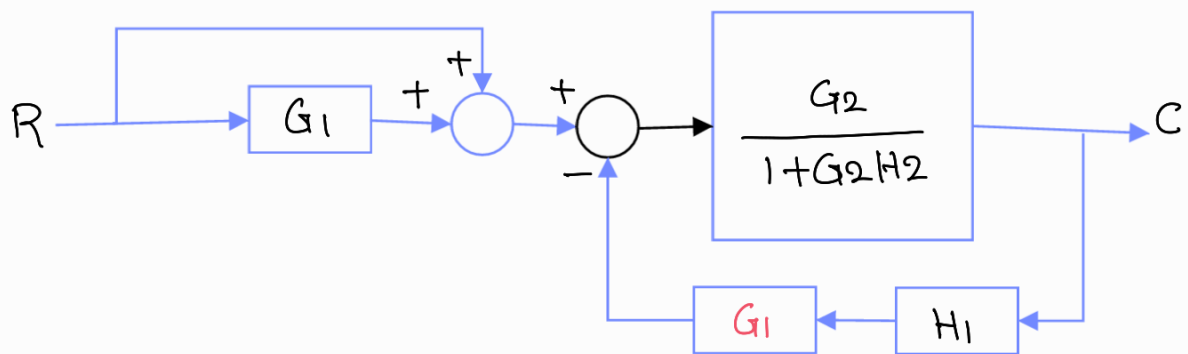
→ eliminating the feedback path:



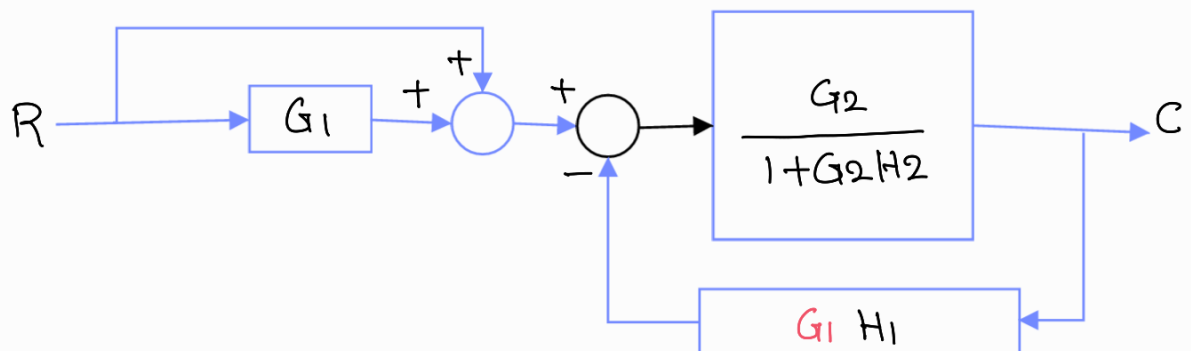
→ moving the summing point after the block:



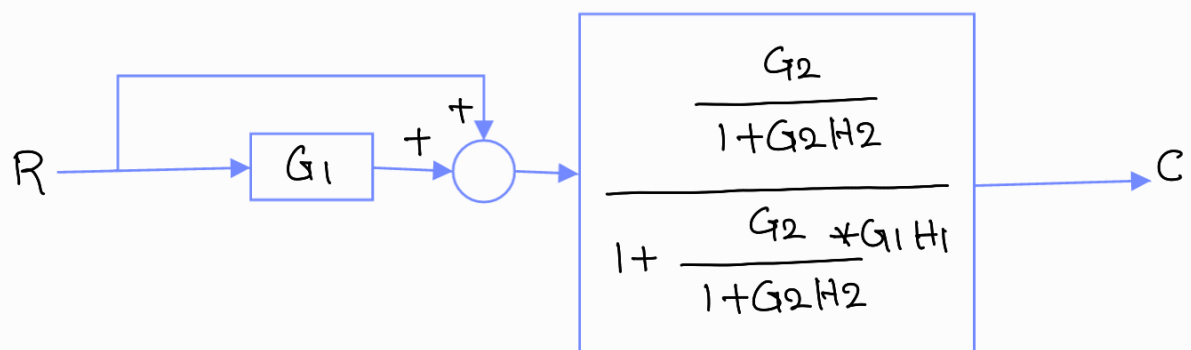
→ interchanging the summing points:



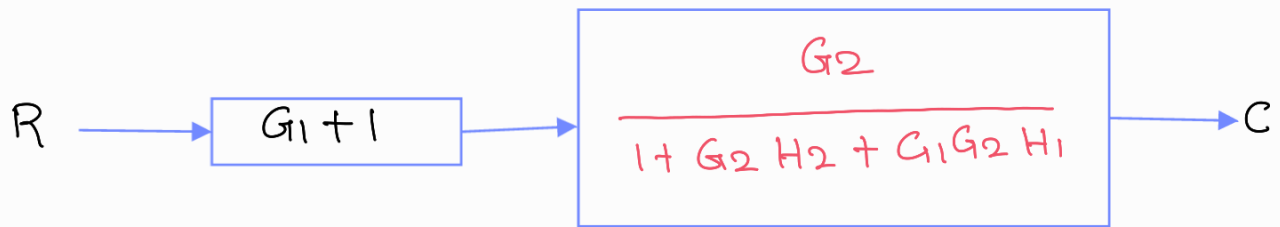
→ combining the blocks in cascade:



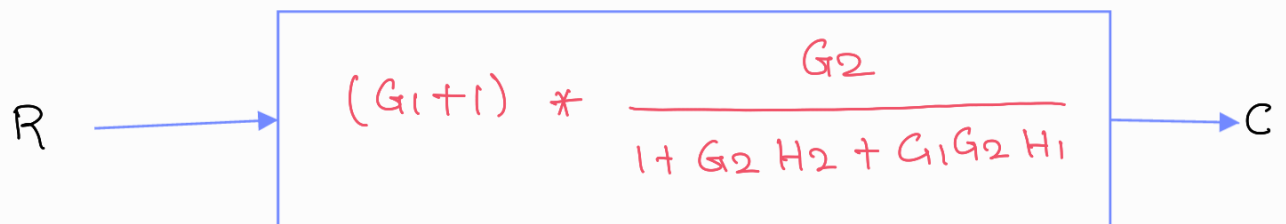
→ eliminating the feedback path:



→ eliminating the feed forward path:



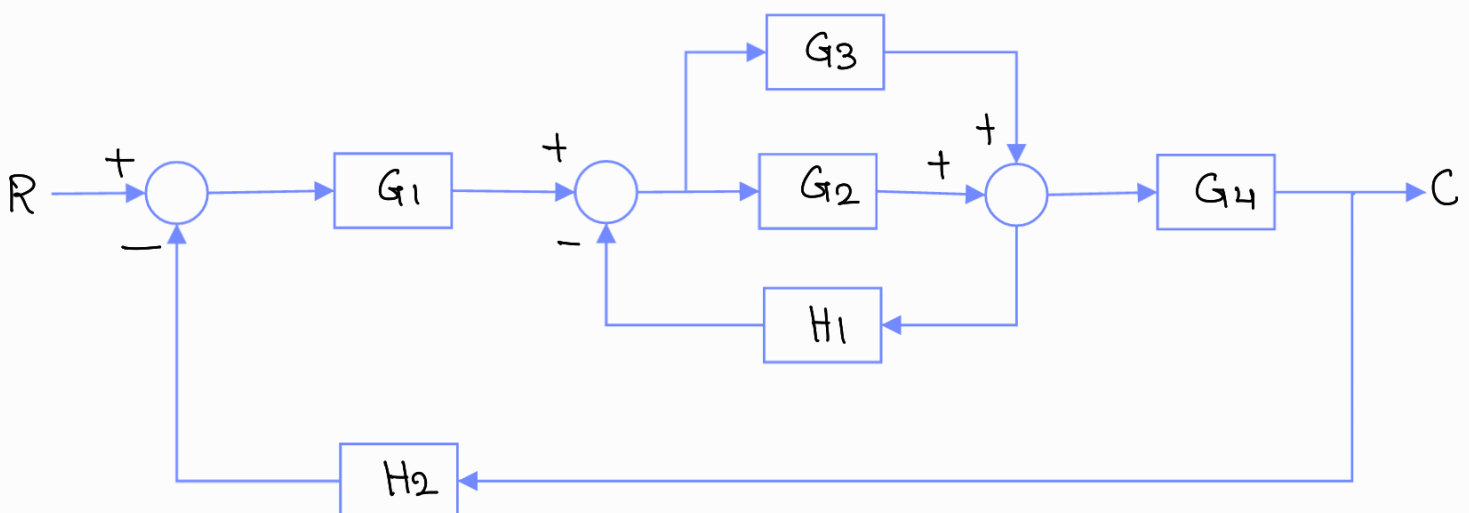
→ combining the blocks in cascade:



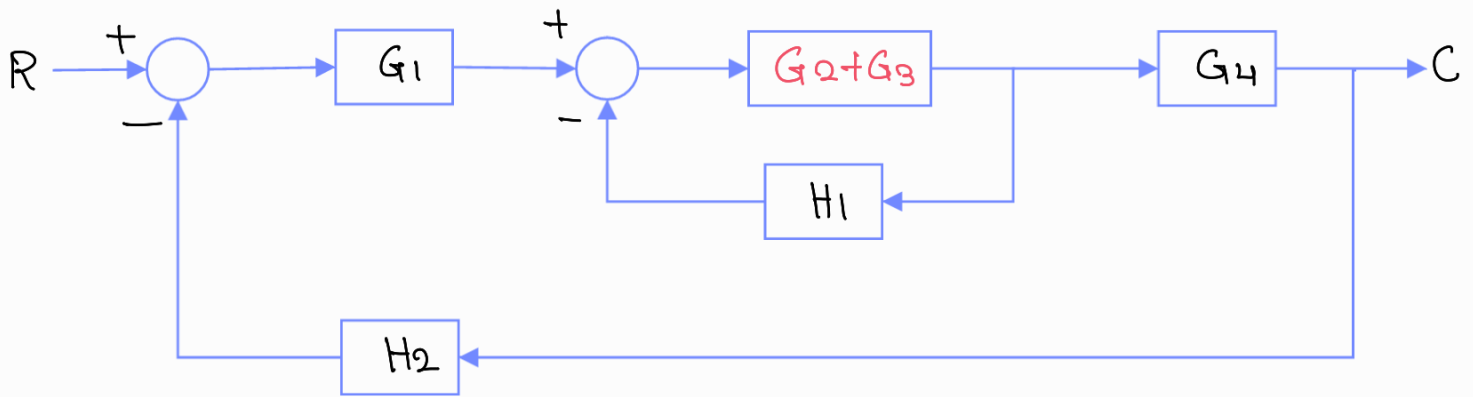
→ overall transfer function:

$$\frac{C}{R} = \frac{G_2 (1 + G_1)}{1 + G_2 H_2 + G_1 G_2 H_1}$$

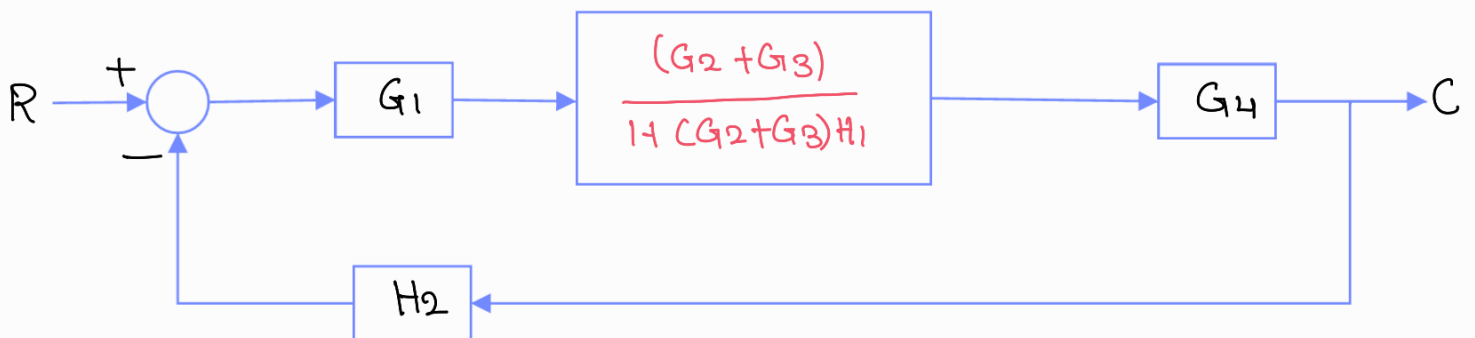
4. Determine the overall transfer function for the given system.



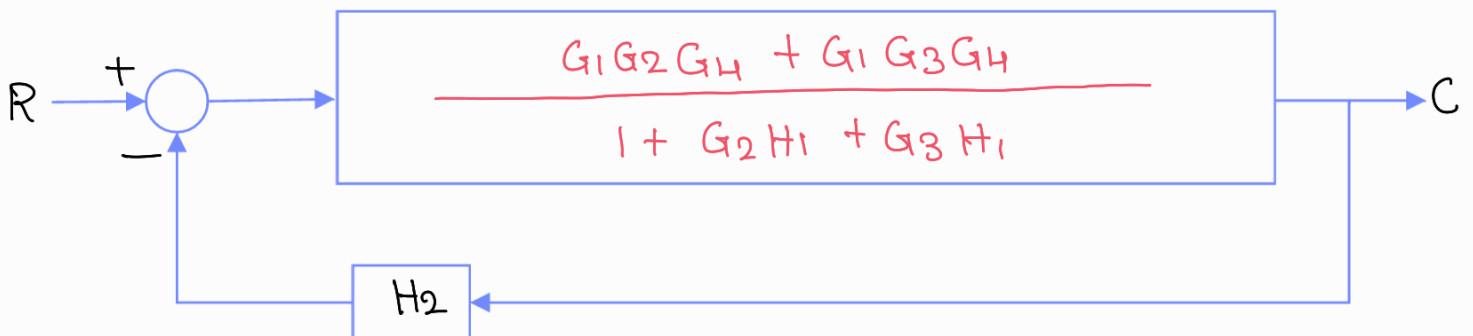
→ combining the parallel blocks:



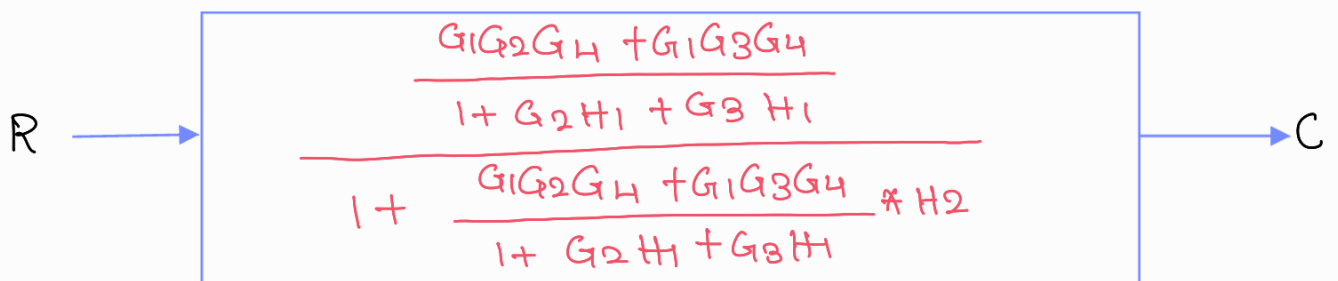
→ eliminating the feedback path:



→ combining the blocks in cascade:



→ eliminating the feedback path:



→ overall transfer function:

$$\frac{C}{R} = \frac{G_1 G_2 G_4 + G_1 G_3 G_4}{1 + G_2 H_1 + G_3 H_1 + G_1 G_2 G_4 H_2 + G_1 G_3 G_4 H_2}$$